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Abstract

Write here a concise summary of your thesis (150–300 words). The abstract should cover: the physical context and motivation, the methods used, the main results obtained, and their significance. Avoid equations and references if possible.

Keywords: keyword one, keyword two, keyword three, keyword four.

Contents

Chapter 1

Introduction

Snow plays a critical role in Earth energy balance,

1.1 Thesis outline

This thesis is structured as follows:

- *Chapter 2* provides the theoretical background needed to understand the work presented in this thesis. The first part focuses on snow physics. It begins with the nucleation of ice crystals in cold clouds and follows the snowflakes down to the ground, where they accumulate to form a new layer. It then describes snow metamorphism and the energy balance of the snowpack, and finally liquid water percolation and runoff generation. The second part reviews snow measurement techniques, distinguishing between manual observations, automatic monitoring networks and remote sensing, and discussing the strengths and limitations of each approach. The third part examines how climate change is affecting snow cover, outlining the current state of snow dynamics worldwide.

- *Chapter 3* introduces the study area and the three main snow products that cover it. It opens with a description of the Italian hydrological portion of the Greater Alpine Region (GAR), which encompasses the Italian Alps and the Northern Apennines. Each dataset is examined in turn, with particular attention to how it is generated and the estimates it provides over Italy. The chapter closes with a comparison of the three snow products, highlighting the discrepancies between their estimates and thereby motivating the need for a new approach to snow monitoring at the national scale.

- *Chapter 4* presents the model used to reconstruct the snow water equivalent (SWE) dataset.

Chapter 2

Snow basics

Snow is a fundamental component of the cryosphere, which encompasses all regions of the Earth where water exists in solid form. Alongside sea ice, glaciers, ice caps and ice sheets, snow cover plays an essential role in regulating Earth's climate by reflecting a significant fraction of the incoming solar radiation, consequently reducing the amount of energy absorbed by the ground. In addition, snow also acts as a thermal insulator due to its low thermal conductivity, protecting the soil and the ecosystem from the abrupt temperature fluctuations typical of winter. Snow is also a key factor in the hydrological cycle: water stored in solid form during winter is gradually released into streamflow through spring and summer, ensuring a continuous water supply and mitigating the severity of certain droughts. At the same time, snow has considerable implications for human activities. For instance, it plays a vital role in the economy of the Alpine region by supporting winter tourism through skiing and other winter sports. Its contribution to runoff is equally important, as it supports irrigation and provides a source of drinking water. Meltwater also provides a substantial contribution to hydropower generation. With the ongoing climate change, these dynamics are expected to shift, making the understanding and modelling of snow related processes increasingly important. Earlier snowmelt and reduced solid water storage are among the most direct consequences of a temperature rise, with potential repercussions for multiple aspects of human civilization. The aim of this chapter is therefore to provide a broad overview of snow dynamics, before focusing on snow measurement techniques, which are essential to assess the state of the snowpack.

2.1 Snow physics and dynamics

The formation of snow is the result of a chain of microphysical processes occurring within clouds, from initial ice nucleation to the eventual fall of snowflakes to the ground.

For the atmosphere to release its water content, whether in liquid or solid form, the air must first reach a state of supersaturation with respect to water or ice. This condition is typically achieved through upward motion of air masses which causes adiabatic cooling of air parcels. The temperature drop strongly reduces air's capacity to hold water vapor, driving the parcel toward, and eventually beyond, saturation (Wallace and Hobbs 2006).

Snowfall can only be produced by clouds that extend above the 0 °C level. Although temperatures within these cold clouds can drop well below freezing point, supercooled water droplets can still coexist with ice particles. The freezing process, known as ice nucleation, occurs through two different mechanisms:

- *homogeneous nucleation*, which occurs in pure water droplets without foreign particles. The first step in the freezing process is the formation of an ice embryo, which must exceed a critical size to produce a decrease in the free energy of the system. If this condition is met, the freezing process continues until the entire water droplet is turned into ice. The average size of the ice embryo increases as temperature decreases, so that below a certain threshold, which is around −40 °C, freezing becomes almost certain. If homogeneous nucleation were the only mechanism responsible for ice particle formation, snowfall would be extremely rare at mid-latitudes. Since this is not the case, an additional mechanism is necessary.

- *heterogeneous nucleation*, which relies on the presence of a foreign particle. Cloud droplets typically contain such impurities acting as freezing nuclei. Their presence promotes the formation of an ice-like structure, thereby enabling freezing at temperatures considerably higher than those required for homogeneous nucleation. It is also possible for foreign particles not to be part of the water droplet itself, but instead to trigger freezing through contact or deposition. Regardless of the specific mechanism involved, the most effective ice nuclei are insoluble in water and have a crystallographic structure similar to ice.

Once nucleated, the ice particle grows through further microphysical processes, whose relative dominance determines its final shape and size. The three main crystal growth mechanisms are:

- *vapor deposition*, which occurs whenever the air is supersaturated with respect to ice. For example, in mixed-phase clouds dominated by supercooled droplets, vapor molecules are deposited onto the ice crystals, causing them to grow significantly faster than the surrounding water droplets. The growth rate reaches a maximum around −14 °C, since at that temperature the difference between the saturated vapor pressure over water and over ice is largest, thus driving vapor molecules more effectively towards ice crystals. Studies on the growth of ice crystals by deposition have shown a strong relationship between crystal shape and the temperature at which the growth process takes place. This strong temperature dependence produces an alternating pattern between plate-like and column-like ice particles, as shown in Figure 2.1.

- *riming*, which occurs exclusively in mixed-phase clouds. Whenever an ice crystal collides with a supercooled droplet, the latter may freeze onto the former, resulting in an increase in ice mass. The collision frequency depends on cloud density and composition, but typically numerous riming events take place, sometimes producing a graupel, namely an ice particle in which the original ice crystal can no longer be identified. Riming can also lead to the formation of hailstones, which typically range from a few millimeters to several centimetres in diameter.

- *aggregation*, caused by collisions between ice particles. The probability of adhesion is strongly linked to crystal shapes and increases with increasing temperature, reaching a maximum at around $-5\,°C$, where ice surfaces become particularly "sticky".

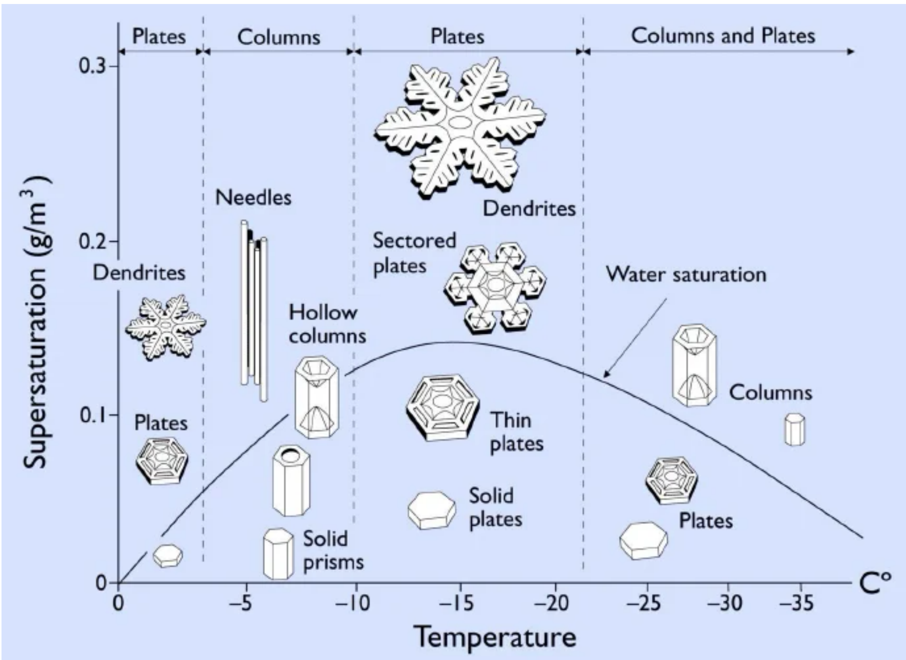

Figure 2.1: The Nakaya diagram illustrates the morphology of ice crystals growing via vapor deposition as a function of cloud's thermodynamic state. Crystal shape is primarily governed by temperature, which determines whether it develops in a planar or columnar structure, while the degree of supersaturation with respect to ice modulates the complexity of the resulting form, with higher supersaturations generally linked to more complex structures. Image taken from Libbrecht 2012.

Once an ice particle grows large enough to overcome updraft forces and turbulent motion within the cloud, it begins to fall toward the ground. The temperature profile encountered by the descending particle ultimately determines the type of precipitation that reaches the surface. For snowfall to occur, the snowflake, composed of up to hundreds of ice crystals, must remain predominantly below freezing and avoid melting completely before reaching the ground. Typically, snowflakes have fall rates of approximately $1.5\,\mathrm{m\,s^{-1}}$. Snowfall amounts are measured either by depth or by snow water equivalent (SWE), defined as the height of the water column resulting from a complete melting of the snow. Snow depth typically increases of approximately $1\,\mathrm{cm\,h^{-1}}$ during snowfall events. Newly fallen snow density ranges from $20\,\mathrm{kg\,m^{-3}}$ to $300\,\mathrm{kg\,m^{-3}}$ depending on wind patterns and weather conditions, with higher temperatures generally leading to higher density values (Armstrong and Brun 2008).

2.1.1 Snow metamorphism and compaction

Accumulation of snowflakes on bare ground or atop older snow results in the formation of a new snow layer. Snow crystals form a solid matrix delimiting pores filled with humid air or liquid water or both, depending on layer temperature. This structure can be clearly appreciated in Figure 2.2, where the coexistence of the solid, liquid and gaseous phases of water is visible. Snow is thus a porous medium, whose properties are closely linked to its texture.

The arrangement of snow crystals evolves over time as a result of thermodynamic interactions among the constituents of the snow layer. This process, known as *snow metamorphism*, is strongly influenced by thermal and meteorological conditions. Since each layer within the snowpack is subject to different burial depths, temperature gradients and moisture presence, it follows a distinct metamorphic pathway. Consequently, the snowpack develops a stratified structure, with individual layers exhibiting different crystal characteristics.

A key factor governing the evolution of ice particles is the presence or absence of liquid water, which motivates the distinction between dry and wet snow metamorphism. The former occurs when the pores are filled only by air, typically saturated with water vapor. The main mechanisms underlying dry snow metamorphism are:

- *equilibrium growth*, driven by crystal shape. Local variations in surface curvature generate corresponding variations in equilibrium vapor pressure, inducing local pressure gradients that drive vapor from convex toward concave (or less convex) regions. Over time this transfer, sustained by sublimation on convex areas and deposition on concave ones, rounds ice grains, driving snow crystals toward a spherical shape.

- *kinetic growth*, related to the presence of a temperature gradient. Because saturation vapor pressure depends strongly on temperature, such a gradient causes vapor to diffuse from the warmest crystals toward the coldest ones. Here too, sublimation on warm crystals and deposition on cold ones sustain the process, which, when the temperature gradient is strong enough, leads to the formation on facets.

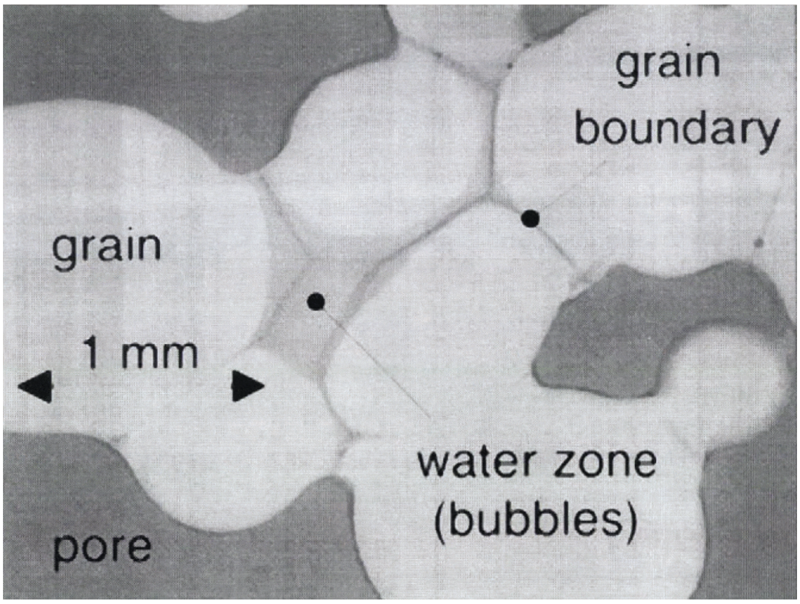

Figure 2.2: Thin section of wet snow under transmitted light, where ice grains delimit pore spaces containing both liquid water and air. The presence of air-filled pores suggest that the sample can potentially retain additional water. Image from Brzoska et al. 1998

These two mechanisms are rarely mutually exclusive: they act simultaneously within the snowpack, with their relative strengths determined mainly by the magnitude of the local temperature gradient compared to the grain's curvature effect. In particular, the kinetic growth of snow particles is a matter of considerable concern in avalanche forecasting, since faceted crystals, if the temperature gradient persists over time, can evolve in depth hoar. This type of layer is notoriously weak and prone to failure, often resulting in large-scale avalanches.

When liquid water is present within the snowpack, the dynamics change substantially: mass exchanges between the three water phases become essential to understand wet snow metamorphism. The main driver of ice particles evolution is, once again, crystal surface curvature, with convex areas associated with lower melting point temperatures. This topological property leads to the melting of the most convex surfaces and to refreezing on the more concave ones, driving ice particles towards the equilibrium spherical shape. The efficiency of this metamorphism channel is linked to the freedom water has to move within the snowpack: at very low water content mass exchanges between the solid and gaseous phases may still dominate.

Regardless of the specific metamorphic process involved, the evolution of ice crystals ultimately results in the compaction of the snow layer. Wind also plays a crucial role in increasing snow density, primarily through the fragmentation of snowflakes upon collision, which allows for a tighter packing of ice particles. Finally, gravity contributes to snow compaction as well, since buried snow layers must withstand an increasing load exerted by the weight of the overlying layers, resulting in higher deformation. The combined effects of metamorphism and deformation strain are typically described through a Newtonian viscosity approach. Snow viscosity depends on several parameters, including temperature, liquid water content and porosity and its values are generally retrieved by means of empirical relations. Overall, the density of older snow layers can easily exceed values of $400\,\mathrm{kg\,m^{-3}}$, sometimes even reaching $500\,\mathrm{kg\,m^{-3}}$ in specific locations, depending on water content and time of the year (Jonas et al. 2009).

2.1.2 Snowpack energy balance

The evolution of a snowpack throughout the season is governed not only by the microphysical processes described above, but also by the exchange of energy between the snow cover, the atmosphere and the underlying ground. Neglecting horizontal energy transfers, the internal energy of the snowpack evolves in time as:

$$\frac{d\mathcal{H}}{dt} = R_N + H_S + H_L + H_P + G, \tag{2.1}$$

where $\mathcal{H}$ is the internal energy per unit area and R_N is the net radiative flux at the surface. H_S and H_L denote, respectively, the turbulent fluxes of sensible and latent heat, governed by the action of turbulent eddies within the boundary layer. H_P accounts for the energy exchange associated with precipitation, since rainfall carries additional heat that must be distributed within the snowpack. Finally, G represents the energy exchange with the ground, which is especially important during the deposition of the first snow layer. Each term in Equation (2.1) is expressed in $\mathrm{W\,m^{-2}}$ and, by convention, a positive sign indicates an energy flux directed into the snow layer.

To fully understand the term R_N, it is essential to consider the dual nature of radiation in the atmosphere. Two distinct sources of radiation can in fact be identified: the Sun, with a surface temperature of around 5800 K, and the Earth-atmosphere system, whose temperature ranges from 200 K up to 350 K. According to the Wien's law, the two blackbody spectra peak respectively at roughly 0.5 µm and 10 µm with negligible overlap, thereby justifying a decoupled treatment of the two spectral domains. Solar radiation is commonly referred to as *shortwave* (SW), while the latter is known as *longwave* (LW). Therefore, the term R_N can be explicitly written as

$$R_N = SW\downarrow - SW\uparrow + LW\downarrow - LW\uparrow, \tag{2.2}$$

where a distinction is made between incoming and outgoing fluxes, as denoted by the direction of the arrows.

Incoming shortwave radiation is subject to absorption and scattering by atmospheric constituents. Consequently, the total SW radiation that reaches the Earth's surface is the sum of a direct beam transmitted through the atmosphere and a diffuse component, resulting from multiple reflection and refraction events. The relative strengths of these two components is strongly dependent on meteorological and seasonal conditions. For example, at the lower solar altitudes typical of winter, the diffuse fraction is larger, since the solar beam must travel through a thicker atmospheric layer, increasing scattering probability. The same occurs at dawn or sunset, when the Sun is close to the horizon. Cloud cover also plays a significant role: under overcast conditions, up to 80% of $SW\downarrow$ can occur as diffuse radiation (DeWalle and Rango 2008). In the shortwave spectral range, snow features an exceptionally high reflectance, thanks to the combination of its microstructure with the optical properties of ice. The ratio between $SW\uparrow$ and $SW\downarrow$, commonly referred to as albedo α, typically exceeds 0.9 for freshly fallen snow, and decreases as impurities are deposited on the snow surface. This parameter is rather important, as it controls the amount of shortwave radiation absorbed, which generally represents the main source of energy for snowmelt. By albedo definition, this quantity can be calculated as

$$SW_{net} = SW\downarrow(1 - \alpha)\,. \tag{2.3}$$

Even at advanced stages of the melt season, snow albedo remains considerably higher than that of the snow-free surroundings. As the snowpack retreats, this bright surface is replaced by darker ground, causing a significant α drop. This results in increased SW absorption, which causes further warming and, in turn, accelerates snowmelt itself in a self-reinforcing loop known as snow-albedo feedback.

Snow behaviour in the thermal infrared range, however, differs substantially from that in the shortwave domain. The snowpack absorbs nearly all incoming longwave radiation, emitted by atmospheric gases, clouds and the surrounding terrain, while radiating according to the Stefan-Boltzmann law and thus behaving as an almost perfect blackbody, with emissivities ranging from 0.97 to 0.99. Longwave radiation exchanges occur both during the day and at night, and are strongly influenced by meteorological conditions. In fact, water vapor significantly absorbs infrared light and, by Kirchhoff's law, is therefore also a strong emitter. As a result, clouds typically increase the downward longwave flux reaching the snow surface, thereby reducing the snowpack's radiative cooling capacity. This effect is especially pronounced at night, when the absence of the shortwave component would otherwise favor significant longwave losses.

Thanks to its relatively low thermal conductivity, the snowpack also shields the ground from rapid atmospheric temperature changes. This property is closely tied to snow microstructure and therefore evolves with metamorphism, as higher snow densities are associated with increased conductivity.

2.1.3 Snow melt and water runoff

The variations in internal energy $\mathcal{H}$, driven by the heat fluxes listed above, ultimately modify the thermal state of the snowpack and determine whether mass is lost through melting or sublimation. The snowpack temperature profile varies on both seasonal and daily scales, typically showing large gradients near the snow surface as a consequence of the low thermal conductivity of snow. The temperature oscillations of the first layers are damped rather quickly with depth, as the thermal state of the deeper bulk remains relatively stable. During winter, snowpacks typically exhibit subfreezing conditions, which poses a significant limitation to runoff generation. In fact, even when the surface temperature rises to 0 °C and the energy balance is positive, the water produced by surface melting refreezes in deeper layers after some percolation. Although this process does not involve any mass loss, it effectively transports latent heat, warming the snowpack and reducing its cold content, which is the energy required to bring the whole snowpack to 0 °C. This isothermal profile is eventually reached and, if a positive energy balance still persists, further melting contributes to filling the pores between ice crystals. This liquid water holding capacity is commonly reported ranging between 3% and 5% of the snowpack mass, although the exact value depends on the snow's porosity and microstructure. Once the water holding capacity is exceeded, the excess water percolates through the snowpack in a process that is mainly gravity driven. Because a typical snowpack is stratified into layers of contrasting grain size and permeability, water flow is rarely uniform as it tends to concentrate along preferential pathways known as flow finger, which are usually triggered by inhomogeneities in microstructure. Typical percolation velocities span more than an order of magnitude, ranging from $1\,\mathrm{cm\,h^{-1}}$ up to $20\,\mathrm{cm\,h^{-1}}$. Ground conditions represent another critical factor in runoff evaluations, since meltwater can in principle infiltrate the soil. If the underlying surface is impermeable, such as frozen ground or bedrock, the excess water instead accumulates and flows downward along the base of the snow cover. If the snowpack is not isothermal throughout, part of this water may still refreeze, further warming the surroundings. The interplay between thermal conditioning, liquid water retention, and preferential flow pathways ultimately governs the timing, volume and rate of snowmelt runoff entering local stream networks.

In high elevation alpine catchments, or at high latitudes, some snow may survive melting through summer. If accumulation persists over multiple years, continued densification ultimately leads to the formation of firn, an intermediate stage between snow and ice characterized by densities ranging from $400\,\mathrm{kg\,m^{-3}}$ to $830\,\mathrm{kg\,m^{-3}}$ and by a porous structure that still permits water percolation. As compaction proceeds further, most pores become sealed, marking the transition to glacier ice proper. The rate of this transformation depends on both accumulation amounts and temperature, since colder climates results in slower crystal evolution. This process directly links seasonal snow cover to the longer lived components of the cryosphere, such as glaciers and ice sheets (Cuffey and Paterson 2010).

2.2 Snow observation and measurement

A detailed knowledge of the snow cover and of its physical properties is a prerequisite for studying snowpack dynamics and for issuing reliable avalanche forecasts. Among the quantities routinely collected during snow measurement campaigns, snow depth (HS) plays a central role: it is defined as the vertical distance between the top of the snowpack and a reference surface, which for seasonal snowpacks is normally the ground. HS should not be confused with snow thickness (DS), which is conventionally the perpendicular distance to the same reference surface. Provided that the slope angle ϕ of the measurement site is known, the two quantities are related by

$$\mathrm{DS} = \mathrm{HS}\cos(\phi), \tag{2.4}$$

so that the two definitions coincide on flat terrain ($\phi = 0$) and progressively diverge as the slope steepens. The geometrical meaning of Equation (2.4) is illustrated in Figure 2.3. Both quantities are expressed in cm and are usually rounded to the

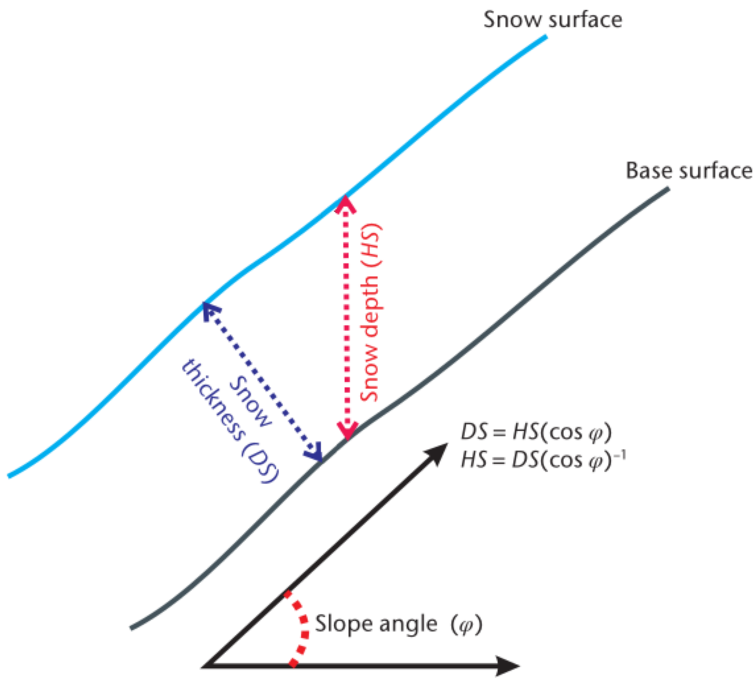

Figure 2.3: Relationship between snow depth (HS) and snow thickness (DS) on a slope. Image adapted from WMO 2024.

nearest integer. While these metrics describe only the geometrical extent of the snowpack, other quantities provide deeper insight into its physical state. The most hydrologically relevant of these is snow water equivalent (SWE), namely the amount of water stored within the snowpack. Being closely related to snow density, it can be expressed as

$$\mathrm{SWE} = \int_0^{\mathrm{HS}} \rho_s(z)\, dz, \tag{2.5}$$

where $\rho_s(z)$ is the vertical density profile of the snowpack. The result is given in $\mathrm{kg}^2\,\mathrm{m}^{-1}$, and can be converted in m (or mm) by dividing by the water density ρ_w. Although formally correct, Equation (2.5) is of limited practical use, since continuous knowledge of the density profile is generally unattainable. What can realistically be measured is the density of the individual layers, so that SWE is more commonly computed as

$$\mathrm{SWE} = \sum_{i=1}^{N} \rho_i(z)\, \Delta z_i, \tag{2.6}$$

where N is the number of layers and Δz_i is the vertical extent of the i-th layer.

Historically, snow metrics were recorded manually, requiring an operator to carry out manual measurements on a daily basis whenever feasible. To ensure continuous data collection even in remote areas, automated weather stations (AWS) have been introduced and are progressively replacing manual observations as well. An automated monitoring network provides a comprehensive overview of snow conditions, but achieving high spatial resolution remains challenging due to high installation costs and maintenance issues. Consequently, certain data can only be obtained through remote sensing, by means of Earth-observing satellites.

While remote sensing provides spatially distributed data across entire regions, manual and automated measurements are taken at specific locations, which must be selected according to strict criteria. To ensure the reliability of these in situ observations, the monitoring site must match the features of the surrounding terrain. In alpine regions, monitoring sites must be free of large boulders and protected from excessive sun exposure, which can cause premature snowmelt and thus yield unrepresentative data. It is equally important to avoid locations prone to preferential snow accumulation driven by local wind patterns (WMO 2024).

2.2.1 Manual snow measurements

Manual snow depth (HS) records are an effective tool for assessing long-term trends, as they can span several decades. Their longevity is due to a straightforward measuring procedure, requiring only a stake or an extendible graduated rod. The snow stake should be vertical, painted white to minimize melt and fixed to the ground before accumulation begins. Snow depth observations should be made from a sufficient distance to avoid any deformation of the snowpack. Because the stake acts as an obstacle to wind and snow redistribution, localized mounds can form around it. Under such circumstances, operator must visually estimate the undisturbed snow surface to recover the actual HS value. For a non-fixed observation carried out with a graduated rod, the snow depth should be evaluated at the same location each time. The observer needs to ensure that the device, manually inserted into the snowpack, reaches the base surface without penetrating the soil, which would result in an overestimate of the actual HS. The rod must be vertical, and an estimate of the slope angle should be paired with every measurement.

Closely linked to snow depth is snowfall depth (HN), which can only be obtained manually. To avoid wind-induced bias this measurement should be carried out in a sheltered location, so that snow accumulates undisturbed on an artificial surface for the prescribed interval (usually 24 hours). Just before clearing the surface, thereby starting the next observation period, a ruler is inserted vertically into the freshly formed layer to obtain a depth measurement.

Although HS and HN define the vertical extension of the snowpack, hydrological applications rely mainly on the snow water equivalent (SWE): the two most common methods for manual measurement are snow pits and snow tubes. A snow pit consists of a hole manually dug into the snowpack down to a reference surface, usually the ground. Initially, the observer performs a stratigraphic analysis to gain deeper insight into the internal structure of the snowpack. The actual SWE measurement is conducted by inserting a sampler of known volume into the exposed snow face; the sample is then weighed to assess snow density. This procedure is usually repeated at multiple depths, either for each identified layer or at fixed spatial intervals, such as every 20 centimetres. This method enables the observer to reconstruct the snow density profile which, when combined with depth information, yields both the layer-by-layer and total snow water equivalent at that specific location. Clearly, the snow pit technique is disruptive and needs to be performed at a different site each time.

If the measurement campaign is focused on a spatial survey rather than a detailed, locally defined inspection, the logical choice is to use a snow tube, as it allows for rapid sampling of the entire snow column to directly retrieve the total SWE, sacrificing layer-specific information. The snow tube must be inserted vertically into the snowpack until it reaches the ground. An estimate of the extracted core volume is made, and the average density is then computed by weighing the snow column. The conversion to SWE is then straightforward. This device should be used where the snow water equivalent is less spatially homogeneous, such at locations with heterogeneous exposure to both wind patterns and solar radiation.

2.2.2 Automatic snow monitoring

Networks of automatic weather station (AWS) require no manual intervention beyond routine maintenance and instrument calibration, and can provide continuous observations even in remote locations where human operation would be impractical. An AWS is typically equipped with a wide range of sensors, such as:

- *anemometer*, which measures wind speed. The blowing wind causes a small windmill, or a structure consisting of three cups mounted on a spindle, to rotate at a rate proportional to wind intensity. This quantity is particularly important, as empirical relations link it to turbulent heat fluxes, thus playing a significant role in energy balance considerations.

- *wind wane*, typically coupled with the anemometer to assess wind direction. It consists of a horizontal arm bearing a flat vertical vane at one end. Incoming wind gusts push against the vain, causing the arm to rotate freely until it aligns with the wind direction.

- *pyranometer*, which measures shortwave solar radiation. An AWS is typically equipped with both and upward and a downward facing pyranometer, in order to quantity incoming and outgoing SW radiation, thus enabling albedo estimates.

- *pyrgeometer*, which measures the longwave (infrared) radiation. Pyrgeometers likewise come in oppositely facing pairs, so that, combined with a pair of pyranometers, an evaluation of the net radiation balance of the surface is possible.
- *barometer*, used to evaluate the atmospheric pressure, a fundamental quantity for characterizing the dynamical state of the atmosphere.
- *thermo-hygrometer*, designed to simultaneously measure both air temperature and relative humidity at a given height above the ground (typically two meters).
- *rain gauge*, consisting of a circular collector and a funnel that channels the collected precipitation into a measuring device. In regions where solid precipitation is possible, the rain gauge may be heated in order to also evaluate the water equivalent of snowfall.

In alpine catchments and at high latitudes, these sensors are typically complemented by additional instruments specifically dedicated to snow monitoring. Mounted together on the same tripod or mounting structure, as shown in Figure 2.4, they constitute what is referred to as an alpine automatic weather station, a key tool for the study of snow related dynamics.

Automated HS measurements can be performed using instruments based on either sonic or optical (laser) technology. Sonic sensors transmit an ultrasonic pulse towards the snow surface and detect the returning echo. Since the speed of sound depends on meteorological conditions, a concurrent temperature reading is required to ensure accuracy. Laser instruments, by contrast, determine the distance to the target surface through interferometric analysis of the returned signal. This approach yields significantly higher precision, but at a grater energy cost.

In both cases, the sensor measures the distance to the upper surface of the snowpack, from which the snow depth is then derived as

$$\text{HS} = \text{Zero depth distance} - \text{Distance to target.} \tag{2.7}$$

The zero depth distance must be determined before snow accumulates and constitutes a key step in instrument calibration. To ensure reliable measurements, the sensor should be positioned well above the maximum expected snow depth, preventing burial and the consequent loss of data during winter. Minimizing interference from

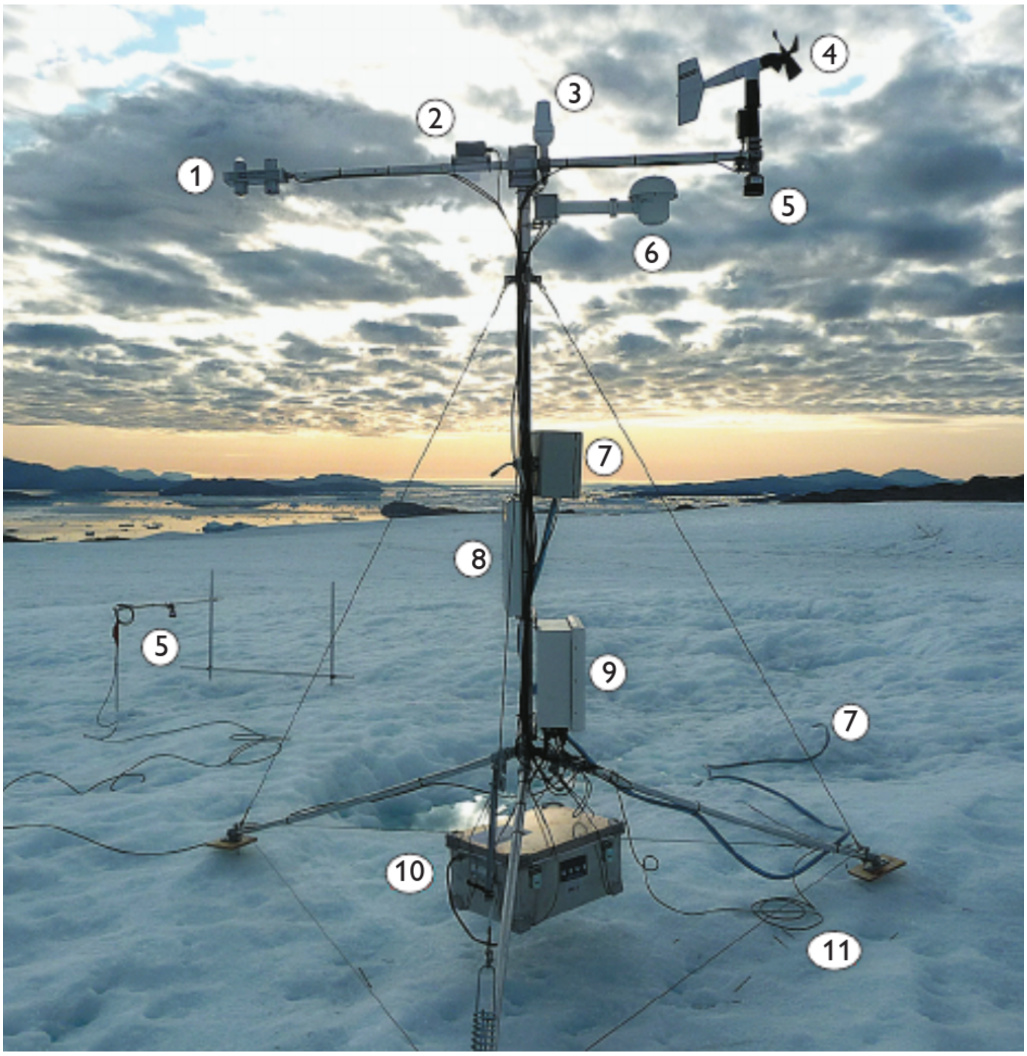

Figure 2.4: The AWS UPE_L which is part of an automated network of the Programme for Monitoring of the Greenland Ice Sheet (PROMICE). The constituents are respectively: (1) radiometer LW + SW, (2) inclinometer, (3) satellite antenna, (4) anemometer and wind vane, (5) snow depth sensor, (6) thermometer and hygrometer, (7) pressure transducer, (8) solar panel, (9) data logger and GPS, (10) battery box and (11) ground temperature sensor. Image adapted from Fausto and As 2012.

the support structure is equally important, since the mast itself acts as a wind obstacle and may induce measurement artefacts. For this reason, HS sensors are typically mounted on a horizontal arm stretching outward from the pole, keeping the target area sufficiently far from the mounting structure.

SWE monitoring can likewise be automated, with the most common technique relying on snow pillows. This device consists of a rubber bladder filled with an antifreeze liquid mixture and connected to a pressure transducer, which yields the SWE estimate. The pillow is installed flush with the ground surface, so that it supports the full weight of the overlying snowpack. As snow accumulates, the resulting increase in hydrostatic pressure is transduced into snow water equivalent values. Snow pillows were first introduced in the late 1960s, and have proved to be a valuable tool for SWE measurements (Beaumont 1965). However, they are now being progressively replaced by snow scales due to the toxicity of the antifreeze fluid, which poses an environmental hazard to the installation site.

2.2.3 Remote sensing

Satellite remote sensing (SRS) serves as a useful tool to fill critical gaps in current knowledge regarding climatic variables, enabling researchers to monitor, usually on a daily basis, even remote locations that would otherwise be inaccessible to a human operator. Furthermore, SRS data substantially improve meteorological reanalysis products, which constitute a cornerstone of contemporary climate change research. Earth observing satellites can be classified into two distinct categories based on their orbital characteristics:

- Polar orbit satellites, which travel roughly from north to south over the poles, targeting a specific longitudinal strip per pass. The entire globe is thus mapped over consecutive orbits. These satellites typically operate at an altitude of approximately 700 km, completing a full revolution in under two hours.
- Geostationary satellites, which occupy a geosynchronous orbit, thereby allowing for real-time meteorological monitoring on a specific target area. Their altitude of approximately 36000 km above Earth's equator results in a significantly lower spatial resolution than that of polar orbit satellites.

The choice of orbit is strictly driven by observational priorities. Consequently, satellites dedicated to cryospheric research favor polar orbit, as maximizing spatial resolution is essential to capture the highly variable distribution of snow coverage across complex terrains.

Satellite-based observations of snow cover extent (SCE) began in 1967, through the efforts of National Oceanic and Atmospheric Administration (NOAA). This monitoring program focused on the Northern Hemisphere, where most of the world's terrestrial snow cover is found, with observations conducted on a weekly basis. Despite the preliminary nature of this work, it highlighted the strong latitudinal dependence of snow cover duration and paved the way for more comprehensive analyses. One of the main limitations of these early observations was their coarse spatial resolution of $\sim 200\,\mathrm{km}$, which was insufficient to adequately characterize snow cover in many mountainous regions (Bormann et al. 2018). A significant advance in sensor technology came with the introduction of the Advanced Very High Resolution Radiometer (AVHRR), which measures Earth radiance in five spectral bands to derive information on temperature, albedo and cloud coverage distribution. The first tests of a spaceborne sensor took place in 1978, but reliable snow cover data have only been available since 1981. This product has been used to estimate snow cover duration and trends since the mid-eighties in various mountainous regions such as the Alps, where SRS data reveal significantly negative trends throughout the 1990s, followed by a partial recovery at the beginning of the new millennium (Hüsler et al. 2014). Although AVHRR offers a substantial improvement in spatial resolution, with pixels of $\sim 1\,\mathrm{km}$, its data reliability is limited by the presence of only one thermal channel for snow-cloud discrimination (Fugazza et al. 2021). To overcome these limitations, the Moderate Resolution Imaging Spectroradiometer (MODIS) was launched in 1999 aboard the Terra satellite, followed by a second unit aboard Aqua in 2002. Both instruments operate in polar orbit, jointly covering the entire globe every one to two days. MODIS is a 36-band spectral sensor spanning from the visible (0.41 µm) to the long-wave infrared (14.4 µm), with a spatial resolution of 500 m. To detect the presence of snow on the ground, the Normalized Difference Snow Index (NDSI) is used, defined as

$$\mathrm{NDSI} = \frac{\mathrm{G} - \mathrm{SWIR}}{\mathrm{G} + \mathrm{SWIR}}, \tag{2.8}$$

where G and SWIR denote the green and shortwave infrared spectral bands respectively. This normalized difference approaches one for a snow-covered pixel, since snow exhibits very high reflectance in the visible portion of the electromagnetic spectrum, while strongly absorbing radiation in the SWIR. An NDSI value greater than zero indicates the presence of snow in the pixel; in practice, a threshold of 0.4, corresponding to roughly 50 % snow coverage, is commonly adopted to classify a pixel as snow covered (Salomonson and Appel 2006).

Although MODIS has proven to be a robust and well-validated tool for characterizing snow cover, the information it provides remains partial, as it is limited to detecting the presence or absence of snow on the ground. In order to assess how much water is stored in solid form, three main spaceborne techniques have been developed:

- *passive microwave radiometry (PMW)*, which is by far the most common method to remotely evaluate SWE. Since the snowpack attenuates the microwaves emitted by the Earth, comparing signal intensities between two frequencies ($\sim 18\,\mathrm{GHz}$, and $\sim 37\,\mathrm{GHz}$) makes it possible to estimate snow water equivalent.

This is the only spaceborne technique that allows for long term studies, as global daily maps have been available since the launch, in 1978, of NASA's Scanning Multichannel Microwave Radiometer (SMMR) (Schilling et al. 2024). Altough the temporal coverage of these series is considerably better than that of other available sources, several factors affect data reliability, such as signal saturation caused by a deep snowpack.

- *gravimetric methods*, which offer a different approach focusing on mass changes. The accumulation of snow on the Earth's surface causes variations in its gravitational field, which can be converted into SWE estimates. This approach is mainly employed by the Gravity Recovery and Climate Experiment (GRACE) mission. Owing to its global coverage, this SWE dataset features a coarse spatial resolution of roughly 300 km, which nonetheless remains useful for continental scale hydrological studies (Schilling et al. 2024).
- *Synthetic Aperture Radar (SAR)*, which consists in the transmission of electromagnetic waves towards Earth and the analyses of the backscattered echoes from the Earth's surface. SAR can penetrate cloud cover and acquire data under all weather conditions. Snowpack properties, such as SWE and snow grain size can be estimated remotely, as longer SAR wavelengths are even capable to penetrate into the snowpack. Furthermore, the phase information recorded by SAR can be exploited to monitor ground deformation and assess the stability of terrain features. Although SAR provides a spatial resolution on the order of tens of metres, its temporal resolution is significantly lower than that of PMW and gravimetric methods, as the revisit time of a specific location is typically five days (Tsai et al. 2019).

2.3 Climate change

Climate change is conventionally defined as an alteration of the climate attributed, directly or indirectly, to human activities that modify the composition of the global atmosphere. This anthropogenic signal is superimposed on the natural variability of the climate system and must therefore be disentangled from a background of spontaneous fluctuations. Over the past few million years, such fluctuations have produced a cyclic alternation between glacial periods, lasting roughly 100 000 years, and warmer interglacial conditions. These climatic oscillations are caused by periodic variations in the orbital parameters governing Earth's revolution around the sun, which generate a weak radiative forcing that is subsequently amplified by snow albedo and greenhouse gas feedbacks (Hays et al. 1976).

The ongoing global warming, however, cannot be attributed to natural variability, as it is primarily driven by the large amounts of greenhouse gases released into the atmosphere by human activities. These gaseous compounds, of which carbon dioxide (CO_2) is the best known, partially absorb the longwave radiation emitted by Earth's surface and re-emit it in all directions, so that only a fraction of this energy escapes, thus causing an increase in surface air temperature. Since the Industrial Revolution the atmospheric CO_2 concentration has risen steadily, from about 278 part per million (ppm) in pre-industrial times to 422.8 ppm in 2024 (NOAA 2025). During the last 60 years, atmospheric CO_2 has risen nearly 100 times faster than during previous natural increases, and present day concentrations exceed those preserved in the ice core record of the last 800 kyr, as visible in Figure 2.5.

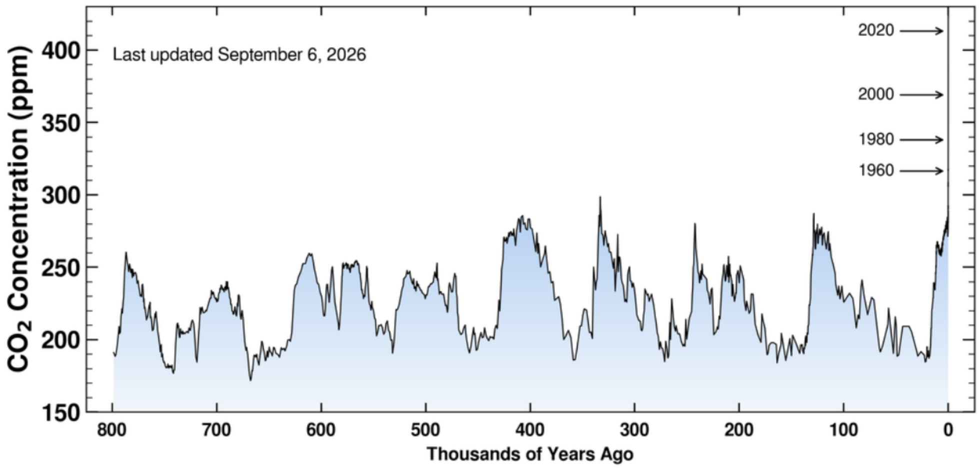

Figure 2.5: Evolution of CO_2 concentration in the atmosphere over the last 800 kyr years. Image adapted from Oceanography 2026.

Relative to the 1850–1900 climatology, the global mean surface air temperature observed over the 2010–2019 period was about 1.1 °C higher, and anthropogenic forcing alone is estimated to account for 0.9–1.3 °C of this increase, with the best estimate being 1.07 °C (Gillett et al. 2021), thus explaining the vast majority of the measured signal. The warming has also accelerated significantly in recent years, as the mean increase rate of roughly 0.18 °C per decade shown during the 1970–2010 period has nearly doubled between 2010 and 2023 (Hansen et al. 2025). This behaviour is clearly visible in Figure 2.6, which shows the time evolution of the global surface temperature anomaly with respect to the 1880–1920 mean.

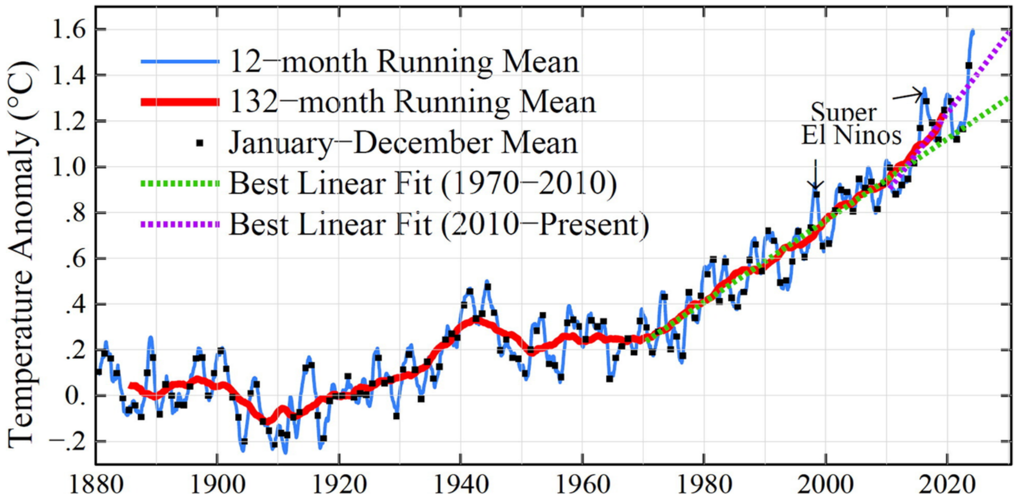

Figure 2.6: Global surface temperatures anomaly relative to the 1880–1920 average. Adapted from Hansen et al. 2025.

2.3.1 Impacts on climate system

The increase in temperatures has significant consequences for every physical component of the climate system. A warmer atmosphere holds more water vapour, which provides additional moisture to storms and thus results in more intense precipitation events. At the same time, warming causes a greater share of precipitation to occur as rainfall instead of snowfall, increasing the risk of flooding in early spring and of droughts in summer, as less water is stored in seasonal snowpacks (Trenberth 2011). Global warming also contributes to sea level rise: global mean sea level increased by 20 cm between 1901 and 2018 as a consequence of the thermal expansion of seawater and of the substantial melting of land based ice masses. The average rise rate nearly tripled over the last century, climbing from 1.3 mm year^{-1} over the 1901–1971 period to roughly 3.7 mm year^{-1} between 2006 and 2018. Furthermore, the continuous warming of the upper ocean layers results in more violent tropical storms and poses an existential threat to maritime ecosystems (IPCC 2023).

Among the many components of the climate system, the cryosphere is arguably the most sensitive to a warming planet, as it can thrive only in cold climates. Over the past four decades, the mean Arctic sea ice extent has been shrinking at an alarming rate of 54 300 km^2 year^{-1}, decreasing from approximately 12.5 million km^2 in 1979 to roughly 10.2 million km^2 in 2021 (Parkinson 2022). This decline is accompanied by a thinning of the ice layer and by a notable shortening of the sea ice season in the southernmost regions of the Arctic, as shown in Figure 2.7. Nowadays, the shores of Siberia remain ice-free for up to three months during summer, and comparable conditions are observed along the coasts of northern Canada. This marked reduction

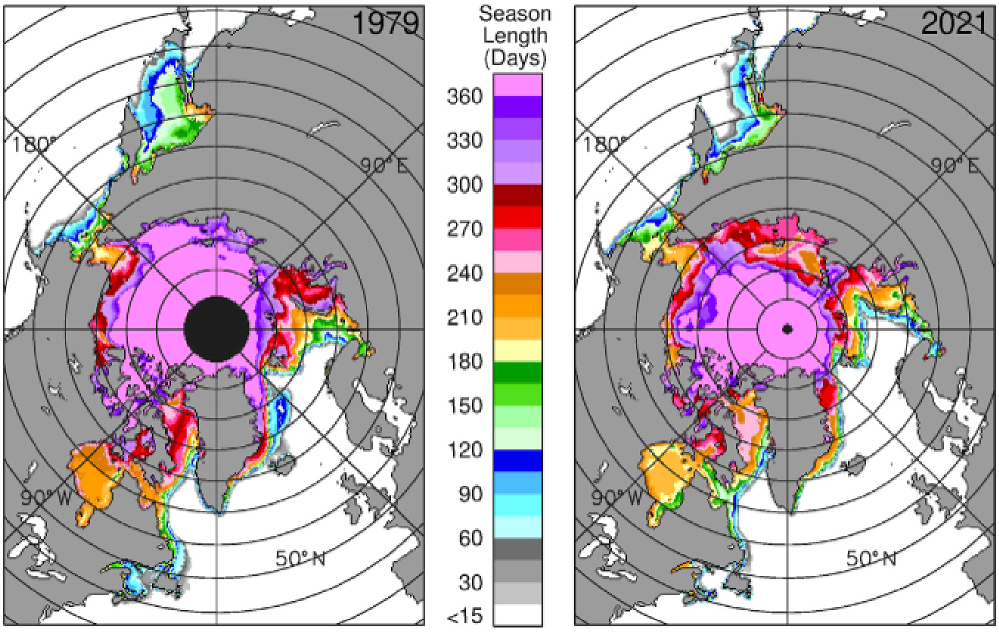

Figure 2.7: Comparison between the sea ice seasons in 1979 and 2021. Image adapted from Parkinson 2022.

in the extent and persistence of sea ice is further worsening the climate crisis through a vigorous ice albedo feedback: as the disappearing sea ice is replaced by open water, which has a significantly lower albedo, more solar radiation is absorbed, leading to additional warming. Ice loss has also picked up in Greenland and Antarctic ice sheets, as the rate of mass loss has more than tripled with respect to the 1990s, surging from 105 Gt year^{-1} between 1992 and 1996 to 372 Gt year^{-1} between 2016 and 2020 (Mudryk et al. 2020). A similar figure emerges for mass loss of glaciers worldwide, which between 2000 and 2023 lost on average 273 ± 16 Gt annually (Zemp et al. 2025). Mountain glaciers are particularly vulnerable to global warning, as rates of environmental change are typically enhanced by elevation. On a global scale, the differences between mountain and lowland trends are 0.21 °C per century for temperature and −11.5 mm per century for total precipitation (Pepin et al. 2025), implying that seasonal snowpacks and glacier ice in these regions will be further affected by a warming climate.

2.3.2 Impacts on the seasonal snow cover

The seasonal snow cover is also strongly affected by the consequences of climate change, as air temperature determines whether precipitation falls in solid or liquid form and regulates the amount of energy available for melt. At the hemispheric scale, snow cover duration (SCD) shows a marked negative trend of −0.44 days year^{-1} (Roessler and Dietz 2023). This decline can be attributed to an earlier onset of spring melt: over the 1967–2026 period, the mean snow covered area in April has been shrinking by 0.43 million km^2 per decade, that is by 1.44 % per decade once normalized to the 1991–2020 mean. This loss rate is significantly larger in June, where the decadal reduction accounts for 14.51% of the corresponding monthly average, or to 1.18 million km^2 (NCEI 2026). Total snow mass also exhibits a negative trend at the hemispheric scale, regardless of the period taken into account. This temporal trends, however, show a significant spatial variability, as peak snow mass over North America is currently declining by -46 Gt per decade, whereas no significant trend is detected over Eurasia, where losses on the European portion are compensated by enhanced snowfall in Siberia. This increase in over Northern Russia is also driven by climate change, as warmer air has a higher water holding capacity causing a shift in precipitation patters (Pulliainen et al. 2020).

A similar picture emerges when focusing on mountainous regions, as aggregate trends over global mountain areas are slightly negative, with an average reduction in snow cover extent of 3.6 ± 2.7 % and a decrease in snow cover duration of 15.6 ± 11.6 days over the 1982–2020 period. Despite the overall reduction at the global scale, the results show substantial spatial variability across mountain ranges and some positive trends driven by climatic forcing may still occur, as visible in Figure 2.8. For instance,

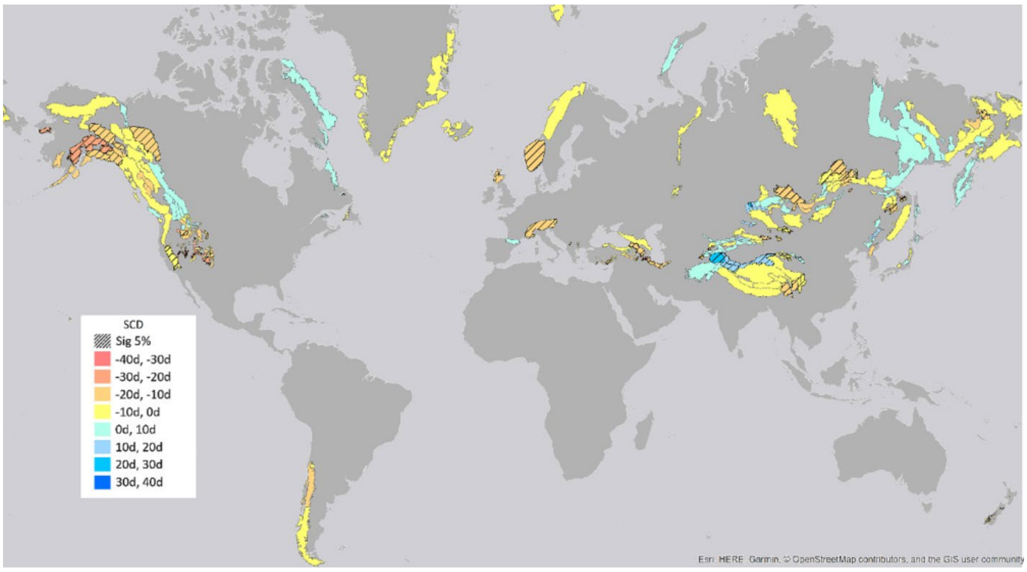

Figure 2.8: Trends in snow cover duration over mountainous terrain between 1982 and 2020. Underlined areas indicate trends that are significant at 95% confidence level. Image adapted from Notarnicola 2022.

the snow cover duration across the Pamir range has been increasing up to 27.4 days between 1982 and 2020 as a result of changes in the winter westerly disturbance, whereas the neighboring Himalayas experienced significant losses (Notarnicola 2022).

Within this heterogeneous picture the European Alps emerge as a particularly critical region. This vulnerability arises from the combined effects of their mid-latitude location and relatively modest elevations, which cause a substantial fraction of the snowpack to remain close to the melting point throughout the snow season. Under such conditions, even a slight increase in air temperature can lead to sudden changes in snowfall and snow ablation patterns.

Chapter 3

Snow products

3.1 Study area

This thesis focuses on the Italian hydrological portion of the Greater Alpine Region (GAR), comprising all Italian territory located north of 43 °N together with the Swiss portion of the Po basin. The main features of this territory are respectively:

- *the Italian Alps*, the southernmost part of the Alpine Arc. They extend for approximately 1200 km, covering an area of about 50 000 km^2. The highest peak, Monte Bianco, reaches 4810 m a.s.l., and summits above 3000 m a.s.l. are fairly common throughout the range. The considerable elevations sustain 901 glaciers, whose total surface area is roughly 368 km^2 (Smiraglia and Diolaiuti 2016).

- *the Po Plain*, a predominantly flat region shaped by an initial phase of marine sedimentation followed by later fluvial deposition. It is drained by the Po river, Italy's longest, which flows for approximately 652 km. Together with the Veneto-Friulian plains, it forms one of the country's principal agricultural and industrial areas, home to roughly 20 million people.

- *the Northern Apennines*, formed by the collision between the western margin of the African Promontory and the Sardinia-Corsica block. Their highest peak reaches only 2165 m a.s.l., insufficient to prevent the complete melt of the seasonal snowpack (Soldati and Marchetti 2018).

The complex orography of the study area, examined in greater detail in Figure 3.1, governs its climate. Indeed, the Italian Alps act as a barrier against cold air masses from the north, while the Apennines shield the Po Valley from the milder influence of the Tyrrhenian Sea. The main climate types found within in the study area are:

- *the Alpine climate*, dominant across the Alps and the Northern Apennines, associated with low temperatures both at night and during winter, and with a summer peak in precipitation.

- *the Po Valley climate*, characterized by cold winters and hot summers, with precipitation peaking in spring and autumn (Ministry for the Environment, Land and Sea 2007).

The high elevations of the Alpine Range, combined with the low winter temperatures, allow the Italian Alps to store substantial volumes of water in solid form, often exceeding 10 Gm^3 (Avanzi, Gabellani, Delogu, Silvestro, Pignone, et al. 2023). Some snowfall also occurs over the Norther Apennines, though in far smaller amounts. These differences in snow cover give rise to distinct hydrological regimes, as up to 50% of the annual stream flow of the main rivers originating in the Italian Alps consists of meltwater (Colombo et al. 2023), and peak discharge is reached in early summer when ablation is most intense. On the other hand, rivers draining the Northern Apennines rely predominantly on rainfall, and often experience low flow, or even run completely dry, during summer.

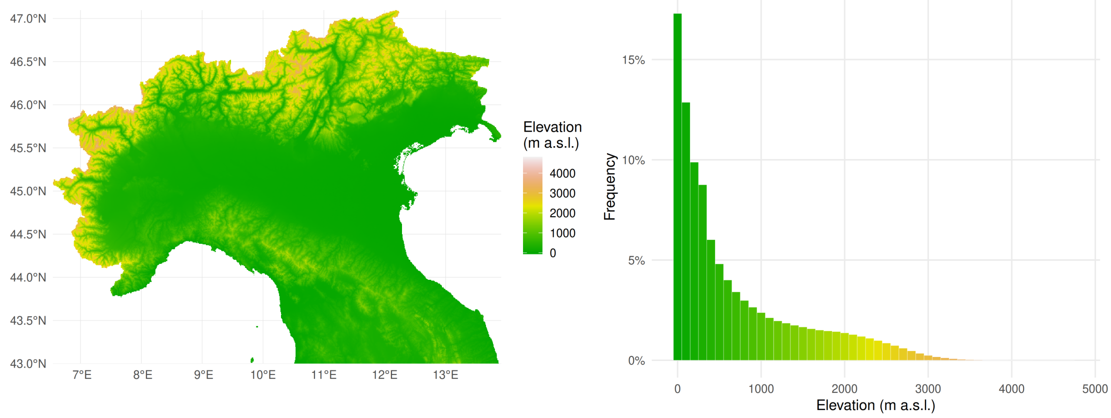

Figure 3.1: Topography and elevation distribution of the study area. The frequency histogram is based on 100 meters wide elevation bins. Elevation data were obtained from ALOS AW3D30 global digital elevation model (DEM).

3.2 MODIS

Thanks to its detailed spectral coverage of the Earth's surface, NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) has become a key tool for mapping snow cover extent and duration across various regions, such as the Greater Alpine Region (Fugazza et al. 2021) or the Karakoram (Almagioni et al. 2025). One of the most recent studies employing MODIS focuses on the Italian territory (Manara et al. 2025), and therefore on the Italian Alps, which constitute the primary focus of this thesis. This dataset, curated by the Department of Enviromental Science and Policy of the University of Milan, provides daily snow cover information at the national scale and, in its most recent version, spans the period from 2000 to 2025. It consists of binary rasters, in which a value of 0 corresponds to bare ground, while a value of 1 denotes a pixel that is at least half covered by snow. The spatial resolution is approximately 500 m.

3.2.1 Dataset generation and gap-filling strategy

To obtain binary rasters describing the presence or absence of snow cover, Manara et al. 2025 proposed an operational processing chain designed to maximise prediction reliability. The procedure makes use of the MODIS Terra Snow Cover Daily L3 Global 500 m SIN Grid V061 (MOD10A1 V61) and MODIS Aqua Snow Cover Daily L3 Global 500 m SIN Grid V061 (MYD10A1 V61) products, both distributed by the US National Snow and Ice Data Center (NSIDC). In these datasets, an integer value is assigned to each grid cell according to the criteria summarised in Table 3.1.

Table 3.1: Integer values in NSIDC snow product.

Value	Condition
0	Snow free
1-100	Snow covered (NDSI value)
200	Missing data
201	No decision
211	Recorded during the nigth
237	Fresh water
239	Sea water
250	Cloud covered
254	Detector saturated
255	Fill value

Cloud cover can substantially reduce data availability across the study area, as the five year mean cloud cover can exceed 80 %. To address the resulting gaps in the MODIS record, Manara et al. 2025 first reduce the complexity of the dataset to three surface classes: snow covered, snow free and missing data. Integer values between 1 and 100 are reclassified as snow covered whenever the corresponding NDSI exceeded the critical threshold of 0.4, and as snow free otherwise. Pixels aquired at night, affected by cloud cover or flagged with an error condition are all labelled as missing data, whereas freshwater pixels are reclassified as missing only when freshwater presence has been detected in less than 10 % of the available observations. Once the number of classes has been reduced to three, Manara et al. 2025 propose a gap filling procedure structured into six sequential phases, each exploiting different spatial and temporal information to progressively reclassify the missing pixels untill the record is complete. These steps, adapted from the previous work of Gafurov and Bárdossy 2009 consist of the following:

- *Terra-Aqua combination*, which exploits the few hour offset between Terra and Aqua overpasses. Since Terra is considered more accurate than its counterpart, the snow map is built primarily from Terra values whenever snow presence or absence is known. If a Terra data is missing, the corresponding Aqua value is examined instead and retained if it is not itself missing. This step assumes that no snowmelt or snowfall occurs between the two overpasses.

- *Short range temporal filling*, which relies on combining snow cover information from adjacent dates to try filling the remaining gaps. For each grid point, the values recorded on the previous and following day are considered: if neither is missing and both agree on the pixel classification, the missing value is reclassified accordingly.

- *Elevation based filling*, which relies on the vertical extent of snow cover on a given day. For each day, the maximum snow line elevation H_s^{max} (above which every pixel is snow covered), and the minimum snow line elevation H_s^{min} (below which every pixel is snow free) are computed. A missing pixel located above H_s^{max} is thus reclassified as snow covered, while one located below H_s^{min} is labelled as snow free.

- *Direct neighbour spatial filling*, during which each missing pixel is compared with its four adjacent pixels. If at least three share the same class, that class is assigned to the missing pixel.

- *Elevation weighted spatial filling*, in which the 3×3 neighbourhood surrounding each missing grid point is examined. If any neighbouring pixel is snow covered and lies at a lower elevation than the central one, the missing pixel is reclassified as snow covered.

- *Pixel-wise temporal filling*, which consists of examining the subsequent values of a missing grid point until a snow covered or snow free classification is found. This value is then assigned to the missing pixel.

After the sixth step, the dataset is entirely free of missing values, providing snow cover information on a daily basis across the whole Italian peninsula.

3.2.2 Results

Following the approach adopted by the authors, particular attention was devoted to temporal snow metrics. In particular, the duration of the snow season (SCD) at a given location was computed as

$$\mathrm{SCD}_{i,j} = \sum_{k=1}^{n} \mathrm{S}_{i,j,k}, \tag{3.1}$$

thereby summing, over an hydrological year spanning from the 1^{st} of October to the 30^{th} of September, the binary values recorded for that specific pixel. In 3.1, k denotes the day of the hydrological year, while $(i,\, j)$ represent the spatial coordinates of the pixel. The SCD is clearly related to the amount of accumulated snow, to temperature and to other climatic drivers. In order to assess whether the enviromental setting induces a shift in the onset or end of the snow season, a fixed date method is typically employed. The snow onset date (SOD) is defined as

$$\mathrm{SOD}_{i,j} = \sum_{k=1}^{D} \mathrm{D} - \mathrm{S}_{i,j,k}, \tag{3.2}$$

where D corresponds to 116, namely the number of days between the beginning of the hydrological year and the 25^{th} of January. Conversely, the snow end date (SED) can be computed as

$$\mathrm{SED}_{i,j} = \sum_{k=D}^{n} \mathrm{D} + \mathrm{S}_{i,j,k}, \tag{3.3}$$

with D retaining the same meaning as in (3.2). These three metrics provide only cumulative information, thus disregarding transient snowfall and meltout events. In the following, only results regarding SCD will be presented: for the analogous analysis concering the onset and the end date of the snow season, the reader is referred to Appendix A.

As expected, MODIS proves capable of capturing the spatial inhomogeneities in snowpack lifetime across the Italian peninsula. This becomes particularly evident in the average SCD map, obtained as the mean of the annual SCD maps and shown in Figure 3.2. Indeed, mountainous regions are associated with the highest average SCD values, typically exceeding two hundred days of snow presence on the ground in high alpine catchments. The orography of the Italian Alps is closely reproduced, as valleys exhibit markedly lower snow cover duration than the surrounding peaks.

The Apennines remain snow free for subsantially longer than the Alps, with only the most prominent massifs, such as Gran Sasso and Maiella, displaying snow behaviour somewhat comparable to their Alpine counterpart. On the other hand, the lowest SCD values are associated with lowland plains, where on average exceeding 5 days of snow on the ground is extrimately rare. SOD and SED exhibit a very similar behaviour to what was previously noted, as mountainous terrain strongly increases the likelihood of a prolonged snow season.

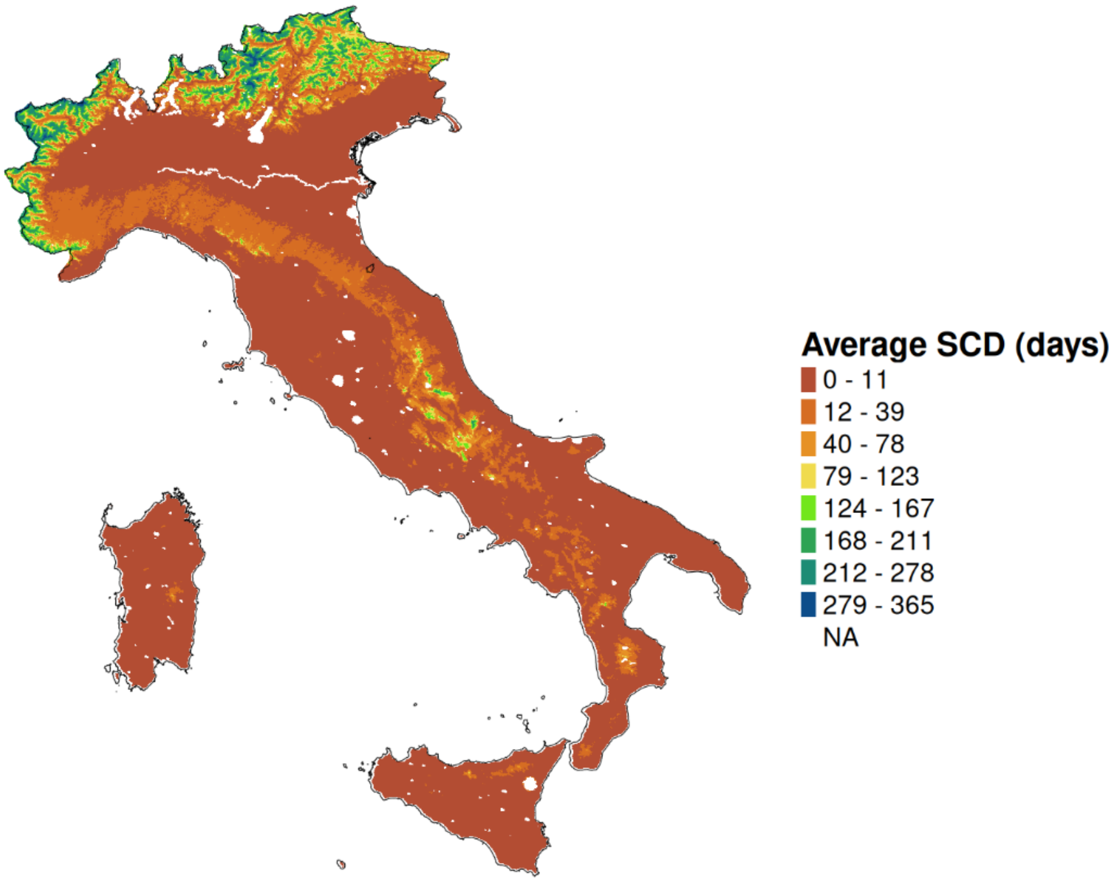

Figure 3.2: Average SCD map across the Italian peninsula for the 2001-2025 period. MODIS accurately reproduces the orography of the Italian Alps, as altitude strongly influence the persistence of snow on the ground. The lowest SCD values are associated with lowland plains, which remain snow free for most of the year.

The analysis of the average SCD map reveals that elevation plays a prominent role in determining both the spatial distribution and the temporal duration of snow cover. To gain deeper insights into this relationship, one possible approach is to divide the Italian territory into 5 m wide elevation bands, in order to plot the quantiles of the SCD distribution as a function of elevation. Notably, the trend shown in Figure 3.3 is clearly monotonic, since higher regions tend to be associated with longer lasting snowpacks, even though a considerable spread is observed within each band. This wide range of possible SCD values indicates that elevation alone cannot fully account for snow distribution: a multilinear regression against several topographic and geographic variables shows that aspect and latitude also provide a non negligible contribution to snowpack behaviour (Manara et al. 2025).

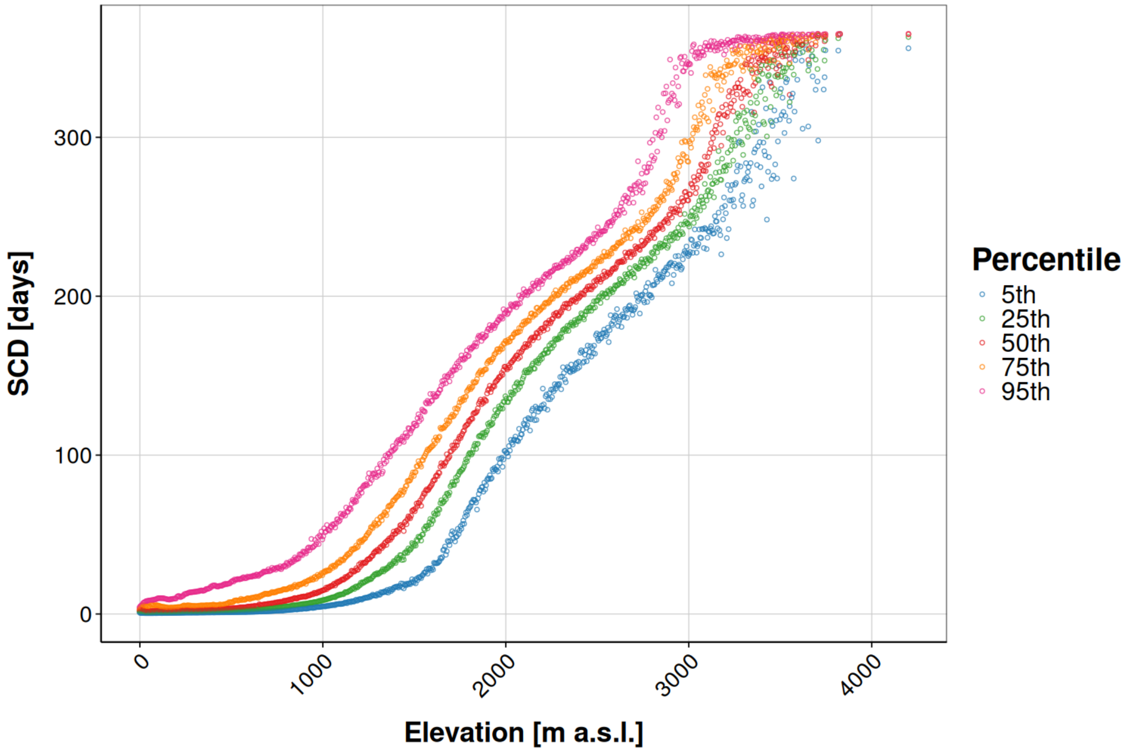

Figure 3.3: Quantiles of the average SCD distribution in each 5 m wide elevation band for the 2001-2025 period. The area taken into account is the whole Italian peninsula. A clear monotonic trend is visible.

The importance of aspect for the duration of the snow season is also evident when examining the spatial distribution of the difference between the SCD value of each individual pixel and the mean SCD per elevation band. In fact, in Figure 3.4 the Alpine valleys are clearly delineated, since south facing slopes experience, on average, a shorter snow season than north facing ones at the same elevation. The Easter Apennines can also be seen to be markedly snowier than the western side of the same ridge. This is attributed to cold continental air masses originating from the Balkan region, which become enriched with moisture as they cross the Adriatic Sea. This water vapor is subsequently released as snowfall through orographic forcing (Capozzi et al. 2025). This east to west gradient progressively weakens toward the Southern Apennines, which typically exhibit a shorter snow cover duration than the other Italian mountain ranges.

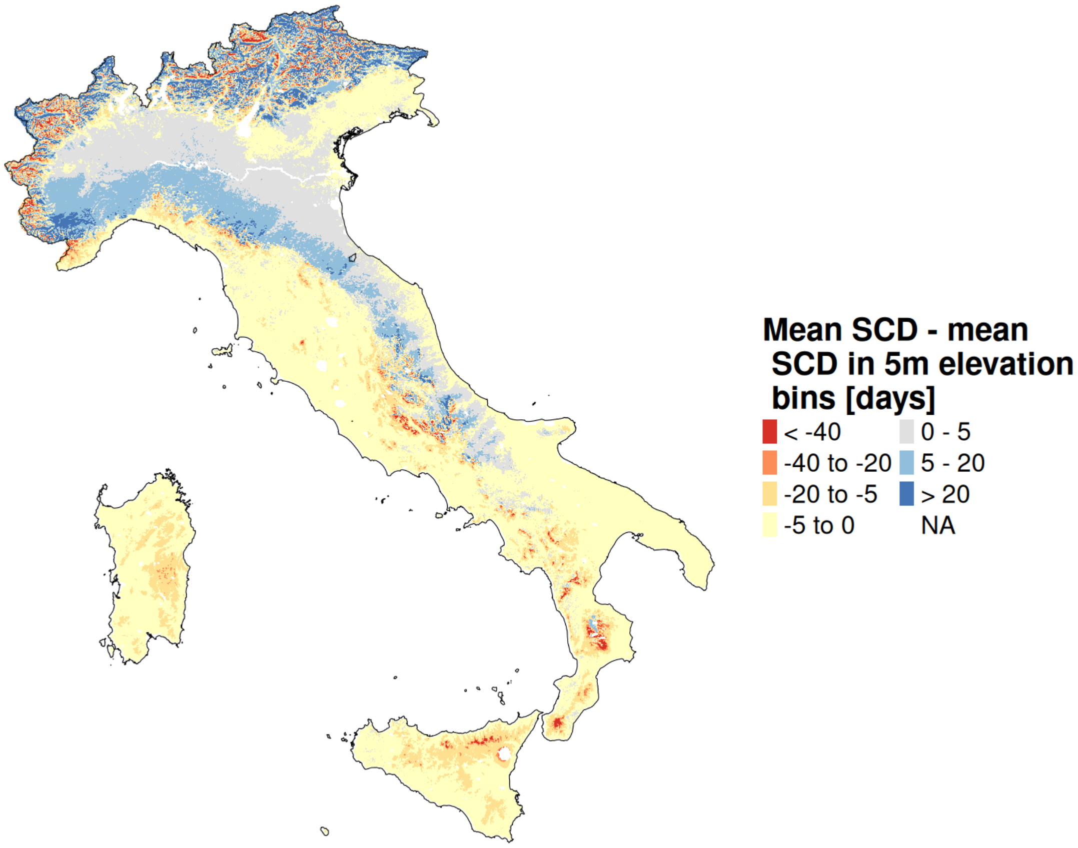

Figure 3.4: Spatial distribution of the difference between the SCD value of each individual pixel and the mean SCD per 5 m wide elevation band.

The snow cover extent and duration observed in any given year cannot, of course, be expected to closely match the 2001-2025 average, since the snowpack is highly sensitive to climatic drivers, such as winter precipitation and temperature patterns, that are themselves subject to considerable interannual variability. This translates into substantial year to year fluctuations in snow cover duration, which can be quantified through the pixel by pixel standard deviation from the mean, toghether with its ratio to the corresponding average SCD value. The spatial distribution of these two quantities is shown in Figure 3.5. The standard deviation exhibits a monotonically increasing relationship with up to approximately 2000 m a.s.l., beyond which it begins to decrease. This behaviour may be partially linked to glacier covered terrain, which, as previously noted, is treated by MODIS as snow covered throughout the whole year, even though bare ice conditions are commonly observed by the end of the melt season. The ratio displays a markedly different pattern, with lower values typically found in high alpine catchments, whereas the highest ones are generally observerd over lowland plains. This trend follows directly from the considerably longer SCD characterising mountainous terrain, which contraints the ratio regardless of the absolute variability.

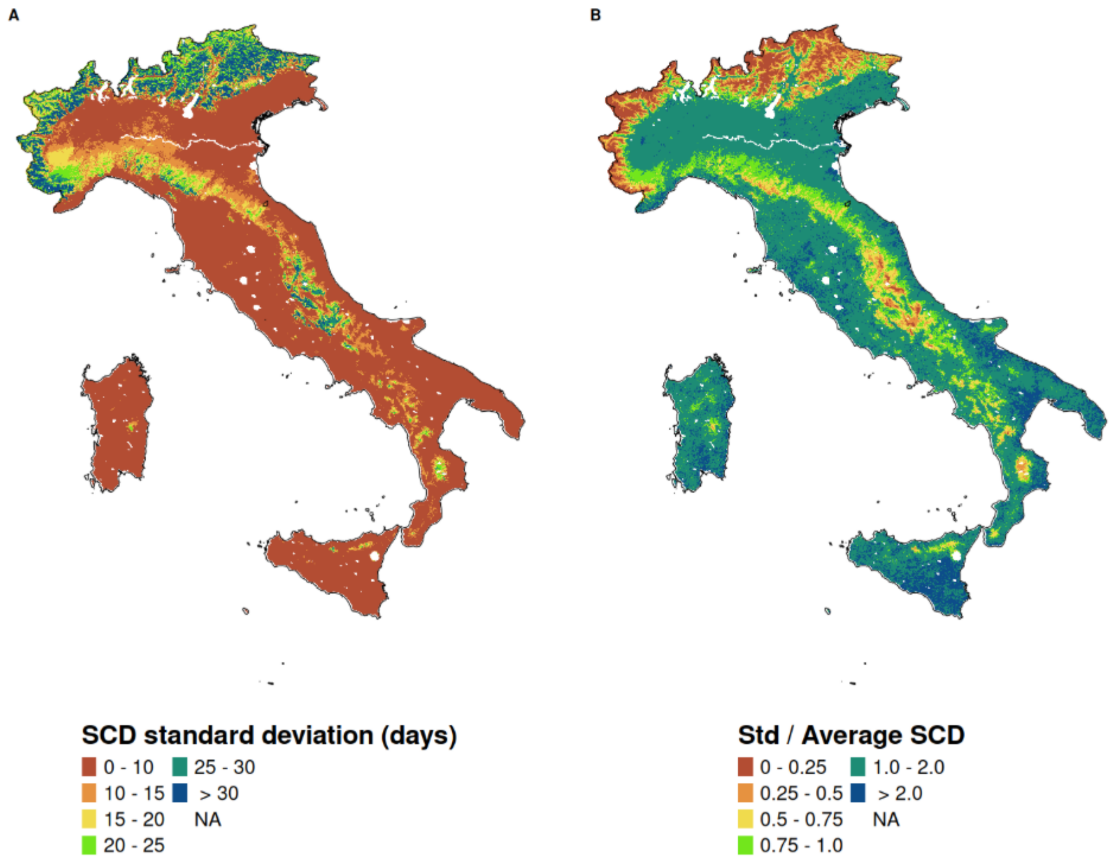

Figure 3.5: (a) Spatial distribution of SCD standard deviation from the mean, computed over the 2001-2025 period. (b) Spatial distribution of the ratio between the SCD standard deviation from the mean and the corresponding mean SCD value for the 2001-2025 period.

3.3 IT-Snow

IT-SNOW is a high resolution, multi-year snow reanalysis dataset covering the entire Italian peninsula at a spatial resolution of roughly 500 m. This dataset consist of daily maps of snow water equivalent (SWE), snow depth (HS), bulk snow density and liquid water content of the snowpack, spanning the period from September 1^{st}, 2010 to August 31^{st}, 2025 in its latest release. IT-SNOW is the output of S3M Italy, a real-time operational monitoring chain developed by CIMA Research Foundation and specifically designed for large scale applications, with the aim of serving as a valuable tool within flood forecasting frameworks. This model achieves a high level of performance by focusing on the key drivers of snow evolution, namely total precipitation, air temperature, relative humidity, and total radiation, thereby providing a spatially consistent reconstruction of snowpack dynamics.

3.3.1 S3M

S3M, also known as *Snow Multidata Mapping and Modelling*, is a hydrology oriented cryospheric model developed to simulate the spatial and temporal evolution of seasonal snowpack. It adopts a raster based approach, in which neighbouring pixels are treated as independent units, so that horizontal mass and energy transport is neglected. All input and output variables are therefore spatially distributed, and the simulated snowpack represents the average conditions within each grid cell. The model provides daily (or even hourly) rasters of snow water equivalent (SWE), snow depth (HS), bulk snow density and liquid water content. From these outputs, the snow cover fraction (SCF) of each pixel can be computed as

$$\mathrm{SCF} = \min\left(1, \frac{\mathrm{SWE}\ [\mathrm{mm}]}{0.1 \cdot \rho_s}\right), \tag{3.4}$$

where ρ_s is the bulk snow density (Kouki et al. 2023). Since a pixel is conventionally regarded as snow covered when at least half of its area is, binary snow extent maps are obtained by assigning a value of one to all grid points with SCF > 0, and zero elsewhere. Beside deepening the knowledge of snowpack properties, this quantity also provides a direct term of comparison with satellite snow cover products such as MODIS.

Within S3M, seasonal snow is partitioned into a dry component, corresponding to the ice matrix, and a wet component, given by the interstitial liquid water. These two constituents are modelled separately, and their mutual interactions govern the overall snowpack dynamics. The total snow water equivalent is thus obtained as

$$\mathrm{SWE} = \mathrm{SWE}_D + \mathrm{SWE}_W, \tag{3.5}$$

so as the sum of its dry (SWE_D) and wet (SWE_W) contributions. The evolution of these two quantities is described by the following mass conservation equations:

$$\frac{d}{dt}(\mathrm{SWE}_D) = S_f - M + R, \tag{3.6}$$

$$\frac{d}{dt}(\mathrm{SWE}_W) = R_f + M - R - O, \tag{3.7}$$

where S_f and R_f denote the snowfall and rainfall mass fluxes, M and R the snowmelt and refreezing rates and O the snowpack runoff mass flux. All the terms in equations (3.6) and (3.7) are expressed in mm w.e. h^{-1} (Avanzi, Gabellani, Delogu, Silvestro, Cremonese, et al. 2022). Solving this system of ordinary differential equations (ODEs), which is decremental to assess snowpack modifications, requires specific parametrizations for each of the mass fluxes involved.

Snowfall and rainfall mass fluxes are estimated from the total precipitation P through a phase partitioning scheme based on air temperature and relative humidity. S_f and R_f are thus computed as

$$\begin{cases} S_f = (1 - p_r)\,P \\ R_f = p_r P \end{cases}, \tag{3.8}$$

where p_r is the rainfall probability, defined as

$$p_r = \frac{1}{1 + e^{\alpha + \beta T_{air} + \gamma RH}}. \tag{3.9}$$

In the previous equation, T_{air} is the air temperature in °C and RH is the relative humidity in percent, while the fixed parameter $\alpha = 22$, $\beta = -2.7$ and $\gamma = -0.2$ are taken from Froidurot et al. 2014.

Snowmelt occurs only when both the current air temperature and its 10-day running mean exceed a threshold T_τ, set to be 1 °C. This dual condition provides a simple way of accounting for the snowpack's cold content, defined as the energy required to warm a subfreezing snowpack until it reaches an isothermal 0 °C state. By requiring sustained mild conditions, rather than a single warm hour, the model avoids triggering unphysical melt events that the snowpack's thermal inertia would not allow. Avanzi, Gabellani, Delogu, Silvestro, Cremonese, et al. 2022 estimate the melt rate through a hybrid temperature index and radiation driven melt parametrization. This formulation improves S3M's ability to reproduce the strong spatial variability that topography induces in snow cover extent and duration, since slope and aspect play a significant role in mountainous terrain that cannot be fully captured by air temperature alone. The hybrid approach adopted in S3M thus reads

$$M = m_{rad}\left[\frac{(1 - \alpha_S)\,S_r}{\rho_W \lambda_f}\right]\Delta t + c_M m_r\,(T_{air} - T_\tau)\,, \tag{3.10}$$

where m_{rad} is a dimensionless coefficient that convert the potential radiative melt into actual melt, α_S is the snow albedo, S_r the incoming shortwave radiation, ρ_W the density of water, λ_f the specific latent heat of fusion and m_r the degree day parametes [mm °C^{-1} h^{-1}]. Both m_{rad} and m_r are calibration based.

In S3M, refreezing occurs whenever the air temperature drops below T_τ and is computed through a degree day approach as

$$R = -c_M m_r \left(T_a ir \; - \; T_\tau\right). \tag{3.11}$$

Since this expression depends solely on air temperature, refreezing is assumed to respond instantaneously to atmospheric cooling, without accounting for the internal structure or thermal gradient of the snowpack.

Ultimately, the runoff O is estimated by simulating liquid water movement through the snowpack using a matrix-flow approximation. This liquid flux is governed by physical properties of the snow cover, as

$$O = \alpha \frac{\rho_W g K_W}{\mu_W}, \tag{3.12}$$

where α is a conversion parameter, g the acceleration due to gravity, K_W the intrinsic permeability of water in snow and μ_W is water dynamic viscosity. This parametrization completes the system of ODEs that, together with those governing the evolution of snow density, are solved at each grid point using the forward Euler method. By default, S3M adopts an hourly integration time step, although this can be adjusted by the user to meet specific application needs.

3.3.2 Results

The IT-SNOW reanalysis provides daily, high-resolution estimates of snow properties, which are a key input for hydrological applications. Over the 2011-2025 period, peak SWE (intended as the total water volume stored in the snowpack) is on average $(11.9 \pm 5.2)\,\mathrm{Gm}^3$, with a minimum of $5.4\,\mathrm{Gm}^3$ in 2022 and a maximum of $23.1\,\mathrm{Gm}^3$ in 2014. This peak is typically reached on 4 March $\pm$ 17 days, with the earliest occurance being 26 January 2023 and the latest 26 March 2013. This means that over the investigated period the maximum of SWE always precedes the 1^{st} of April, which is the date commonly adopted in literature as a reference for peak SWE. The daily evolution of the total SWE volume, shown in Figure 3.6, exhibits great variability driven by thermal and meteorological conditions. The last few years stand out for a markedly reduced water volume stored in the Italian snowpack, with the partial exception of 2024, which nonetheless remained a slightly below average snow year. This is consistent with the persistent snow drought that characterised the early 2020s, driven by the combined effects of reduced precipitation and higher temperatures (Colombo et al. 2023).

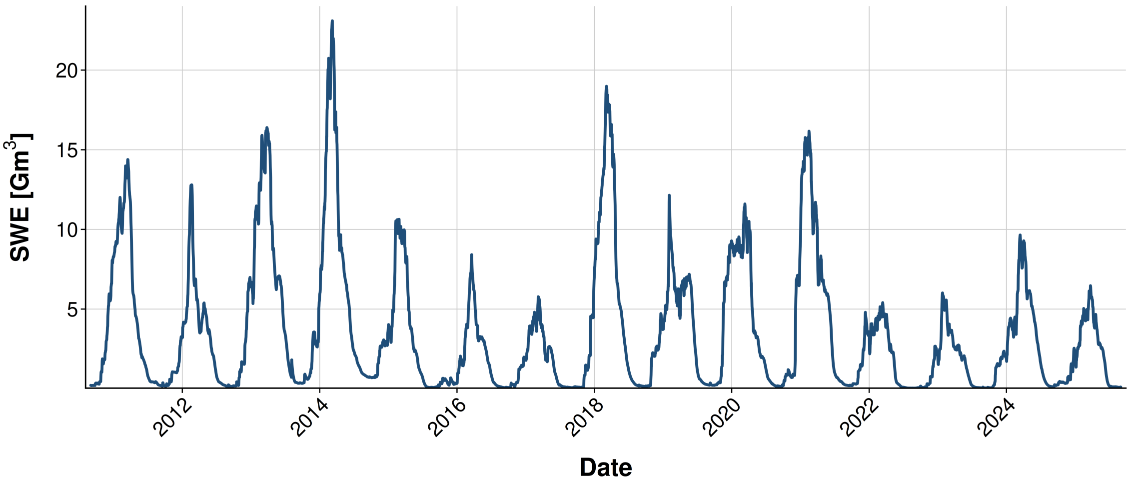

Figure 3.6: Daily evolution of the total water volume stored in the Italian snowpack, computed as snow water equivalent (SWE) integrated over the whole national domain, from 1 September 2010 to 31 August 2025. Data are from the IT-SNOW reanalysis (Avanzi, Gabellani, Delogu, Silvestro, Pignone, et al. 2023).

Besides this strong temporal variability, the modelled snow cover also displays a marked spatial heterogeneity across Italy, with mountain regions experiencing longer and more intense snow seasons. Both panels of Figure 3.7 closely reproduce the orography of the main ranges, showing that the physically based S3M model captures the control exerted by topographic factors such as elevation and aspect on snowpack amount and persistence. According to IT-SNOW estimates, almost 87 % of the mean national winter SWE is stored within the five main Alpine river basins, namely Po River, Adige, Piave, Tagliamento and Brenta, whereas the Central Apennines account for about 7% of total SWE. The remaining fraction is distributed across the rest of the country (Avanzi, Gabellani, Delogu, Silvestro, Pignone, et al. 2023).

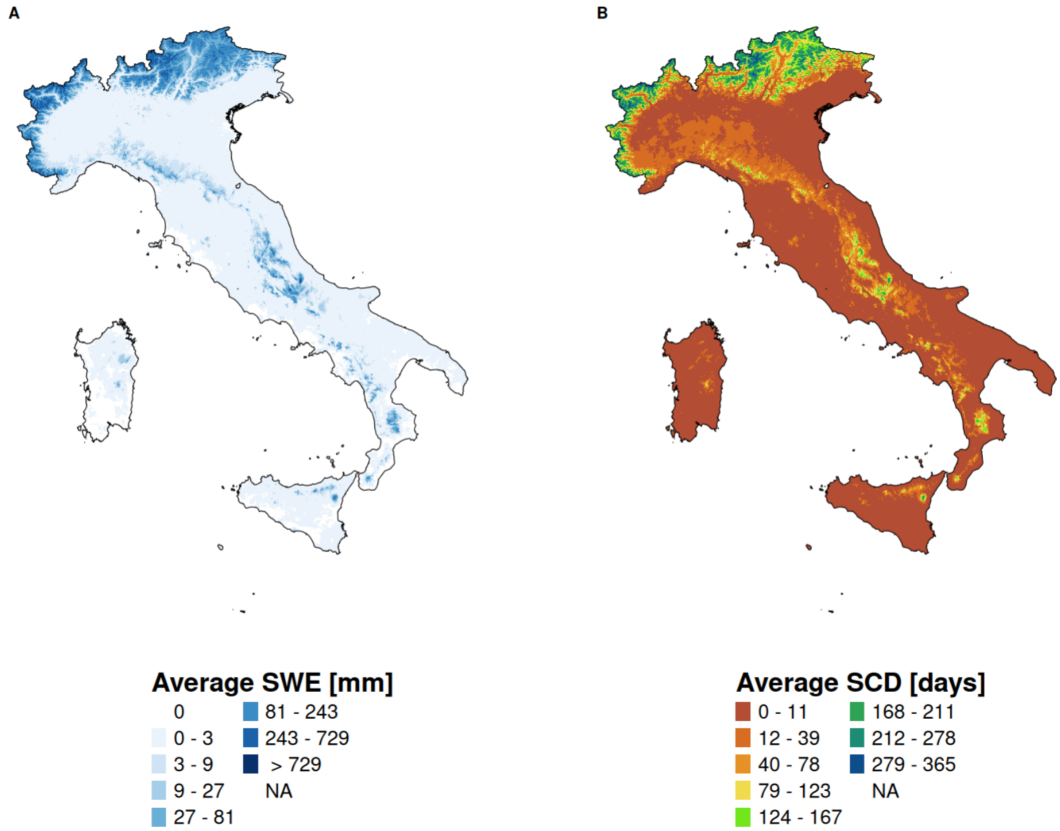

Figure 3.7: (a) Average seasonal snow water equivalent (SWE), computed as the mean of the November–June averages of each hydrological year over the period 1 September 2010–31 August 2025. (b) Average snow cover duration (SCD), computed as the mean of the annual SCD maps across the same period.

3.4 Po River Basin

This snow dataset focuses on the mountainous part of the Po River District, which is an over-regional entity resulting from the union of the Po River Basin with some smaller ones such as Conca and Reno basins. This territory, spanning eight regions of Northern Italy and parts of Switzerland and France, covers almost 87000 km^2. Assessing the amount of water stored as snow in the highlands of this region is particularly important, because the Po River Basin is home to roughly one-third of Italy's total population and produces almost 55% of Italy's hydropower. This snow product is a thirty years long SWE reconstruction, covering a period spanning from 1992 to 2021. For every hydrological year, SWE maps for the mountainous part of the Po River District are given on a daily basis during the period spanning from October 3^{rd} to July $1^{s}t$, which roughly corresponds to the period where snow coverage extends beside glacerized area. These SWE maps have a spatial resolution of approximately 500 m and have been created using a hybrid modelling approach integrating the physically based GEOtop model with in situ snow heigth and snow extent information. This operational chain is referred to as "J-snow".

3.4.1 GEOtop

$$todo \tag{3.13}$$

3.4.2 Results

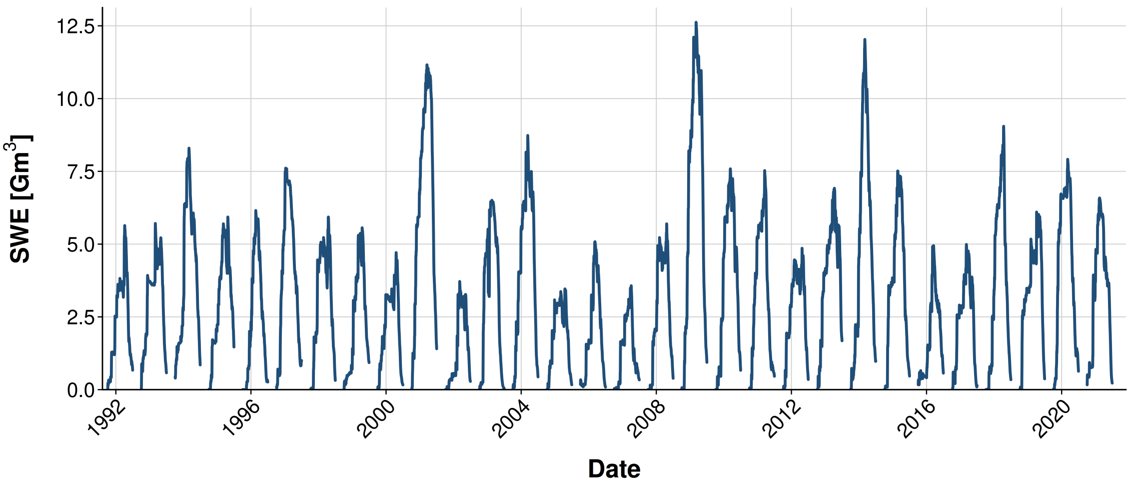

Figure 3.8: MODIS

3.5 Comparison

The snow datasets presented above differ from one another in several respects. ITSNOW and the Po Basin product are the outputs of two distinct physically based model, and both focus primarily on snow water equivalent, consistently with their hydrological purpose. ITSNOW, however, provides richer information than the latter, as it also includes maps of snow density, snow depth and liquid water content, which allow the user to gain a deeper insight into the state of the snowpack. MODIS, in contrast, describes snow cover alone through binary rasters. This discrepancy does not limit the possibility of a comparison, as SWE maps can be converted into binary rasters by means of (3.4) and (3.13). Since these datasets cover different domains, are provided at different spatial resolutions and in different coordinate reference systems, and span different periods, the comparisons reported hereafter are carried out over the largest common overlap, both in space and in time. For the sake of

Table 3.2: Main characteristics of each snow product.

Dataset	Data type	Start date	End date	Study area	Spatial resolution	Reference system
MODIS	Snow cover	24-02-2000	30-09-2025	Italy	0.00417 °	WGS 84 (EPSG:4326)
ITSNOW	SWE	01-09-2010	31-08-2025	Italy	0.00506 °	WGS 84 (EPSG:4326)
Po Basin	SWE	03-10-1991	01-07-2021	Po Basin	500 m	ETRS89 (EPSG:3035)

clarity, Table 3.2 summarizes the main characteristics of each snow product, thereby clarifying the rationale behind some of the comparison choices.

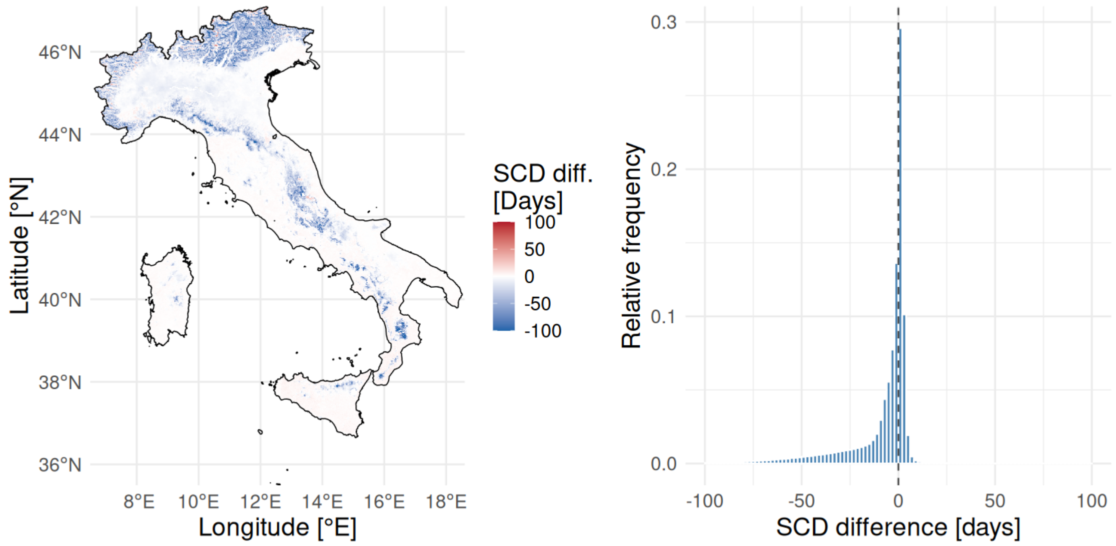

Figure 3.9: MODIS

Chapter 4

SWE model

4.1 Precipitation data

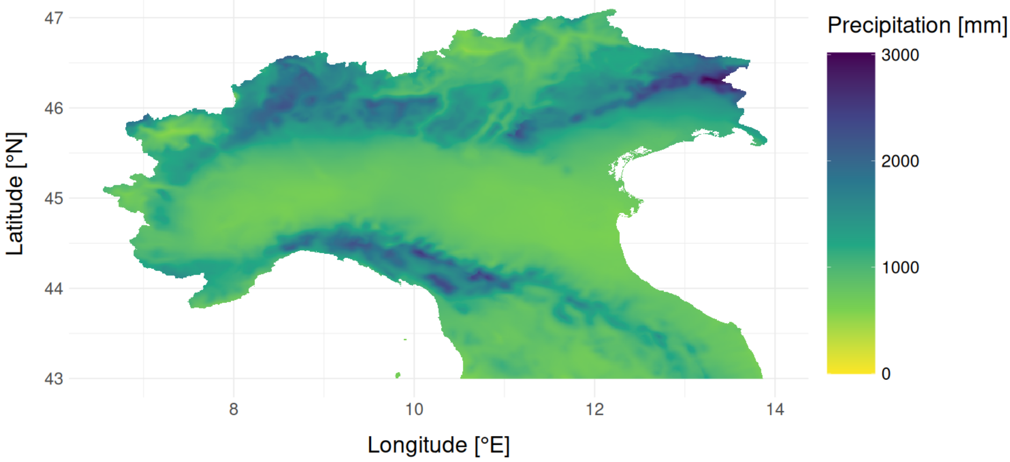

Figure 4.1

4.2 Temperature data

4.3 Phase partitioning and snowmelt

$$P_{s,i} = \begin{cases} P_i & T_{\max,i} < T_{th} \\ P_i \dfrac{T_{th} - T_{\min,i}}{T_{\max,i} - T_{\min,i}} & T_{\min,i} < T_{th} < T_{\max,i} \\ 0 & T_{\min,i} > T_{th} \end{cases} \tag{4.1}$$

$$\mathrm{DDF}_i = \mathrm{DDF}_{ave} + \mathrm{DDF}_{amp} \cdot \sin\left(\frac{2\pi i}{366}\right) \tag{4.2}$$

$$\mathrm{M}_i = \mathrm{DDF}_i \left\{ (T_{mean,i} - T_{th,m}) + \ln\left[1 + \exp\left(-\frac{T_{mean,i} - T_{th,m}}{m_m}\right)\right]\right\} \tag{4.3}$$

$$\mathrm{SWE}_i = \mathrm{SWE}_{i-1} + P_{s,i} - M_i \tag{4.4}$$

Appendix A

MODIS

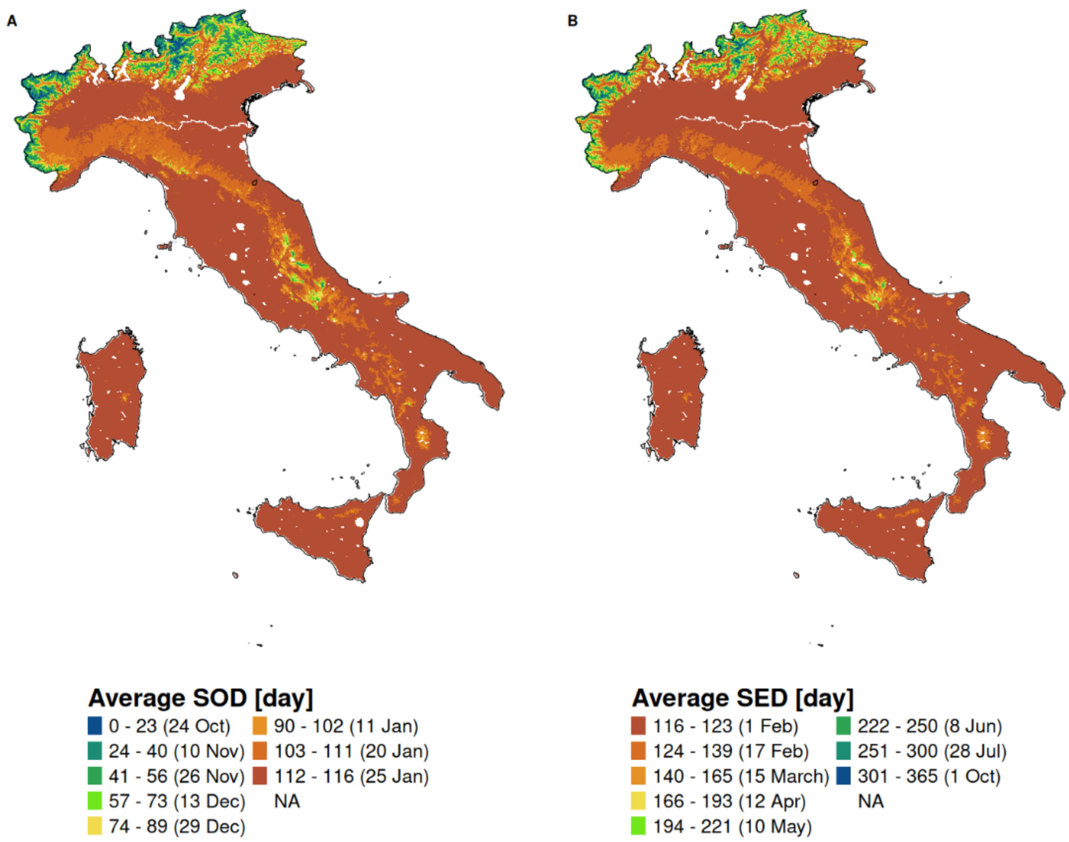

Figure A.1: Spatial distribution of the average Snow Onset Date (SOD, panel a) and Snow End Date (SED, panel b) across the Italian peninsula for the 2001-2025 period. As expected, mountain ranges are associated with an earlier onset and a later end of the snow season.

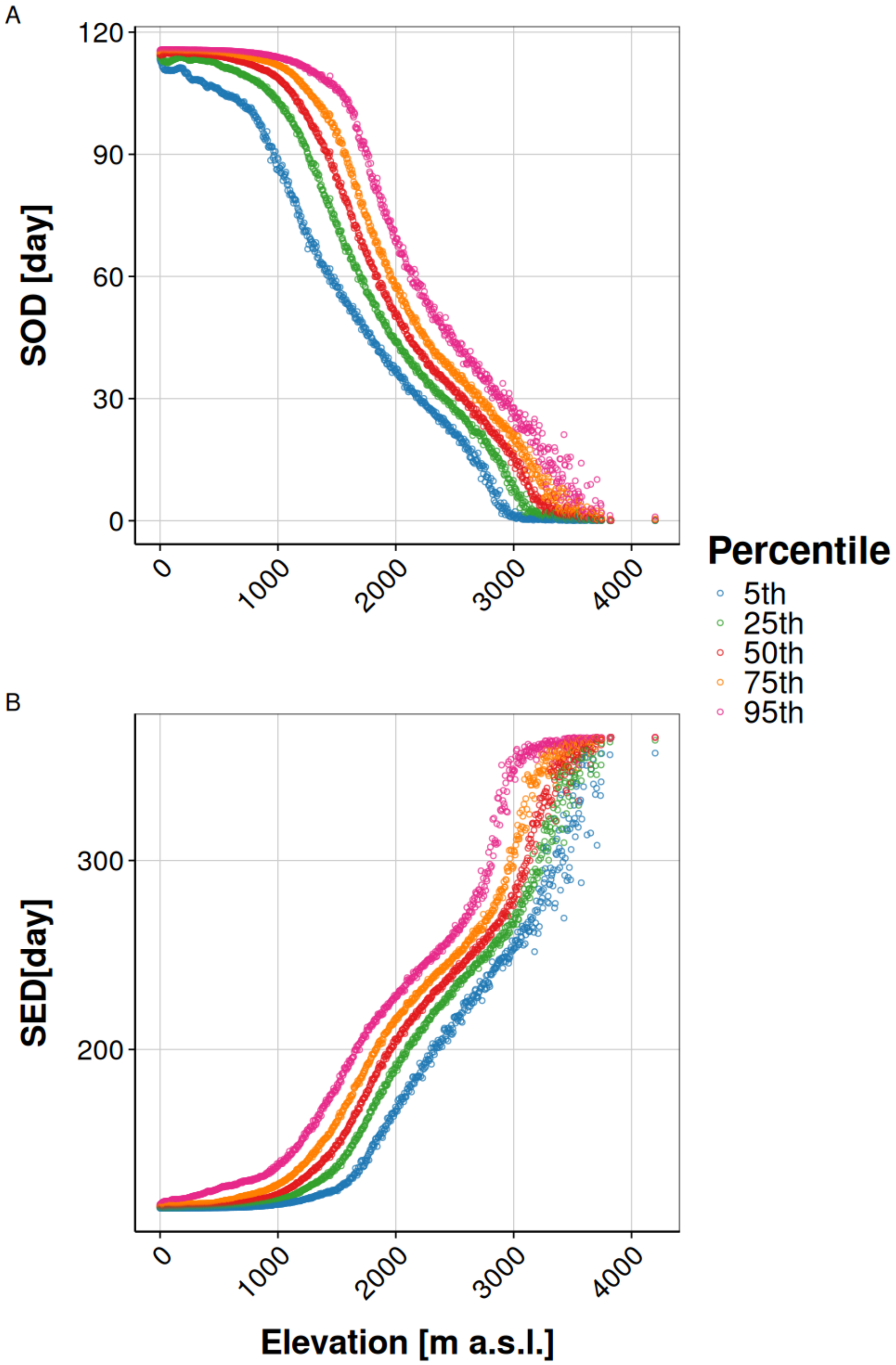

Figure A.2: Quantiles of the average Snow Onset Date (SOD, panel a) and Snow End Date (SED, panel b) distributions within each 5 m wide elevation band, for the 2001-2025 period. Both panels exhibit a monotonic trend with elevation, as high altitude alpine catchments experience earlier onset and later end of the snow season.

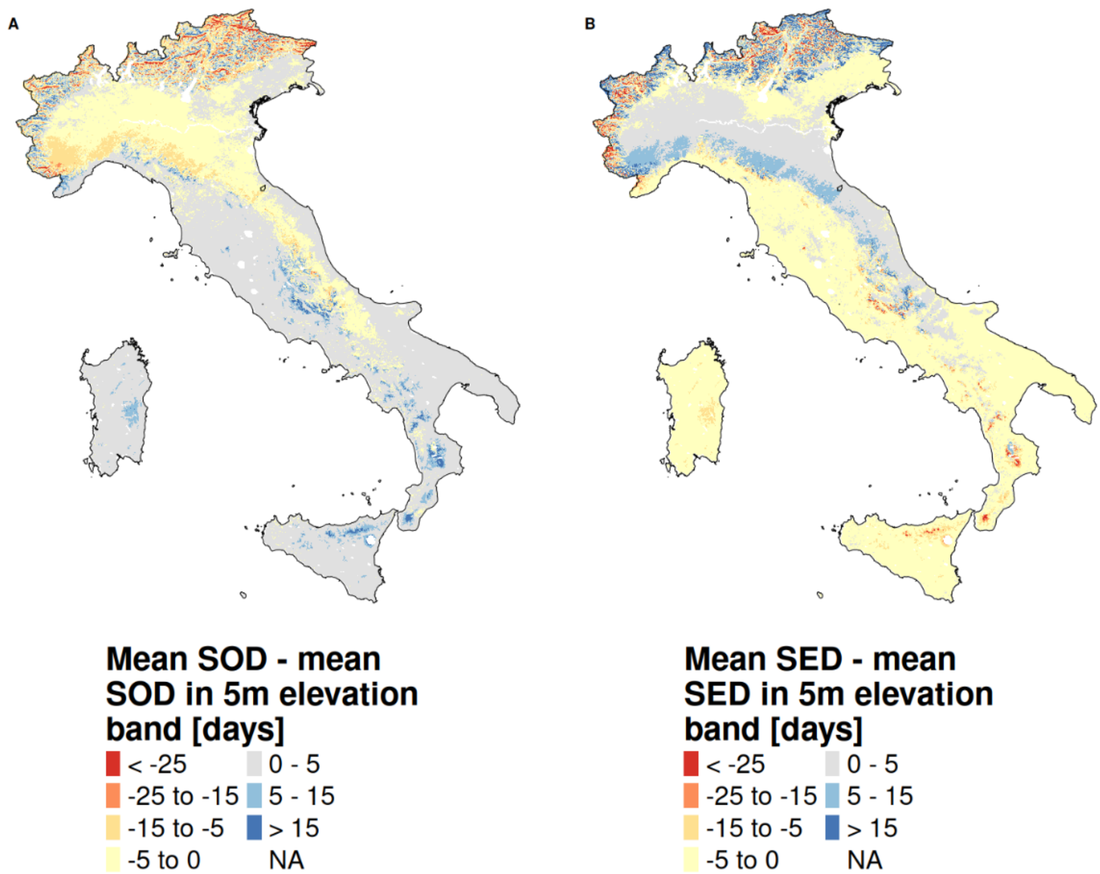

Figure A.3: (a) Spatial distribution of the difference between the SOD value of each individual pixel and the mean SOD of the corresponding 5 m wide elevation band. (b) Spatial distribution of the difference between the SED value of each individual pixel and the mean SED of the corresponding 5 m wide elevation band. The Eastern Apennines typically experience an earlier onset and a later end of the snow season relative to the elevation wise Italian average. This behaviour could be relater to cold air masses originating from the Balkans. In the Alps, pixel aspect plays a significant role, as south facing slopes often experience earlier melt.

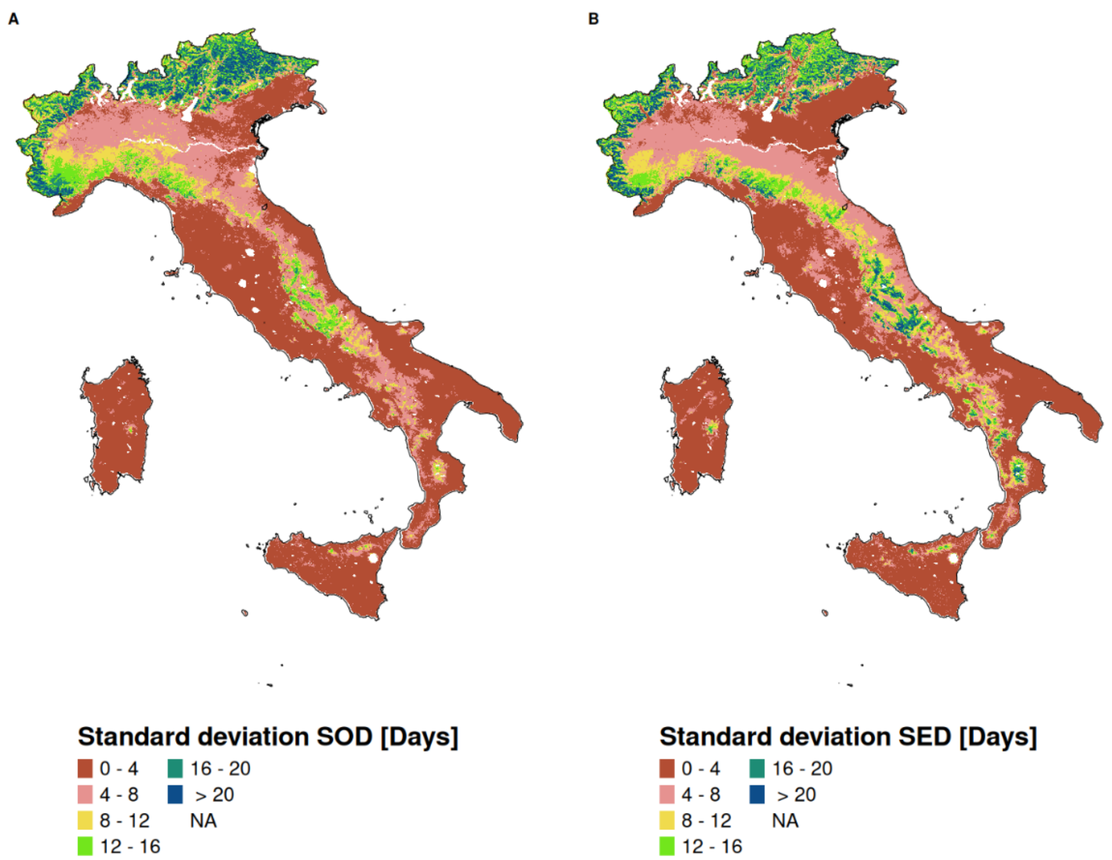

Figure A.4: Spatial distribution of Snow Onset Date (SOD, panel a) and Snow End Date (SED, panel b) standard deviation from the average value computed over the 2001-2025 period.

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